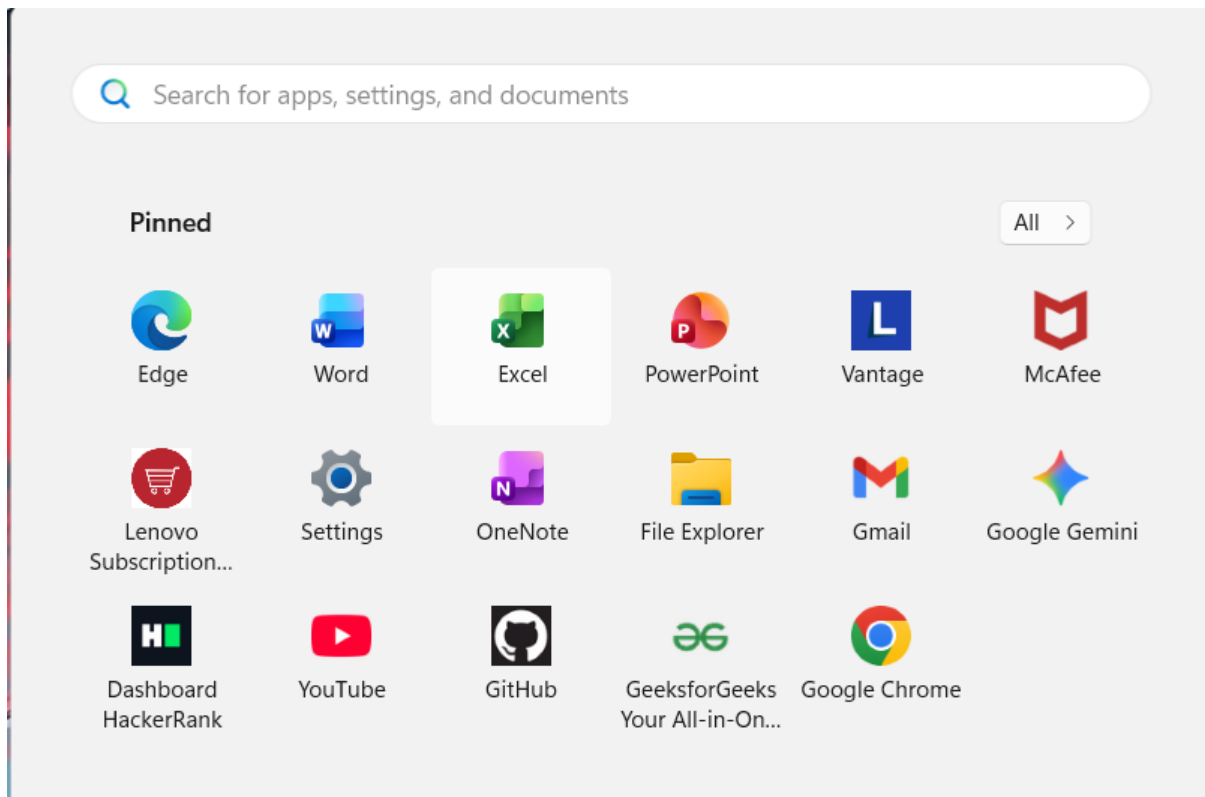


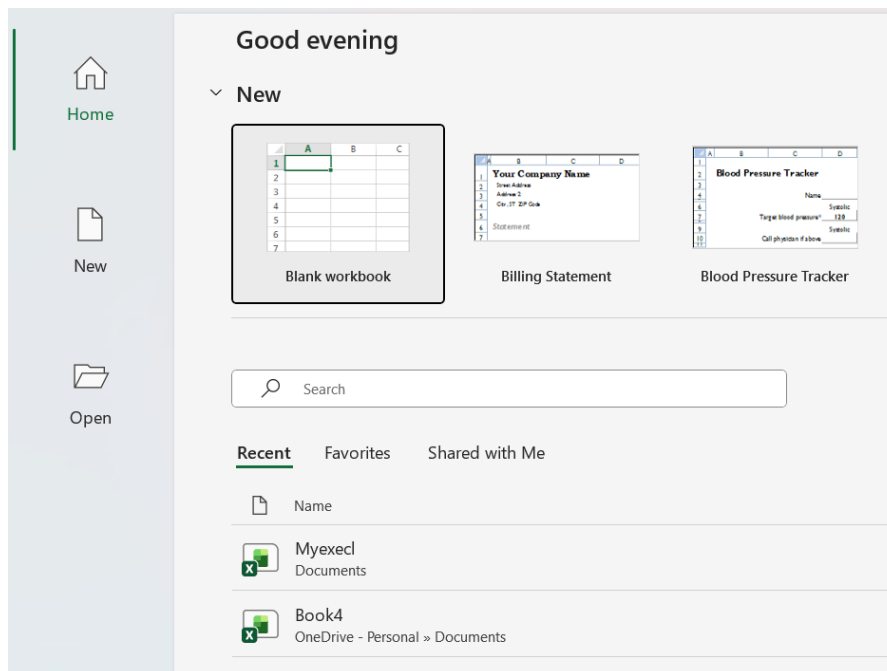
EXPERIMENT NO 1: Getting Started with Excel

1. Creation of the spread sheet

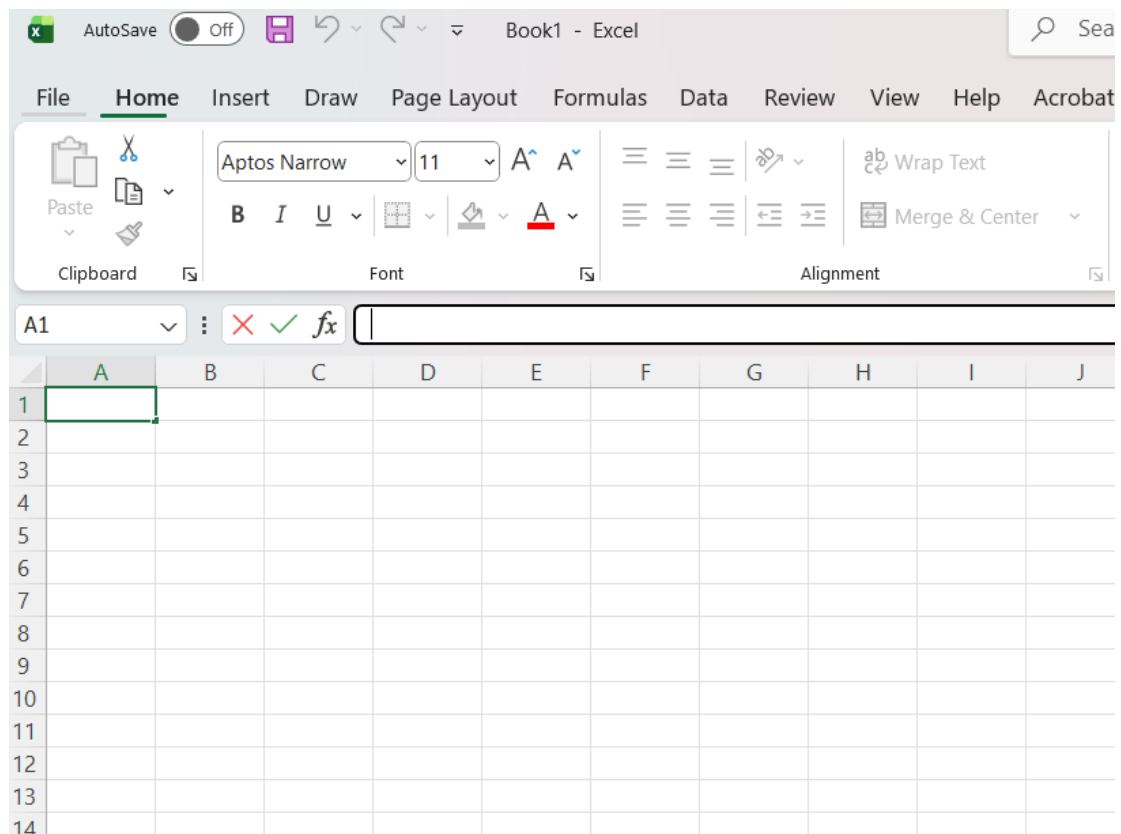
a. Open Excel



b. Select blank workbook

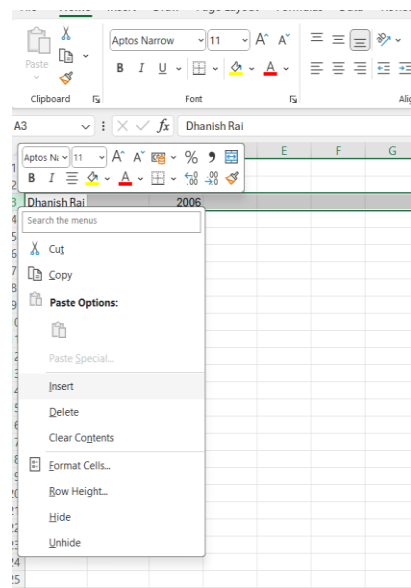


b. Columns and rows headers:

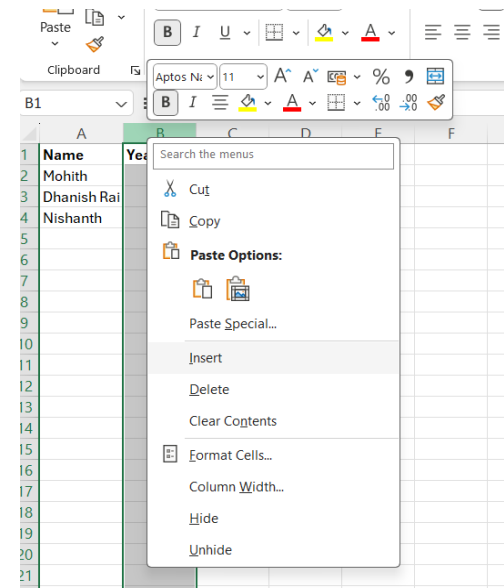


2. Insertion of rows and columns

a. Select the row and the inset option

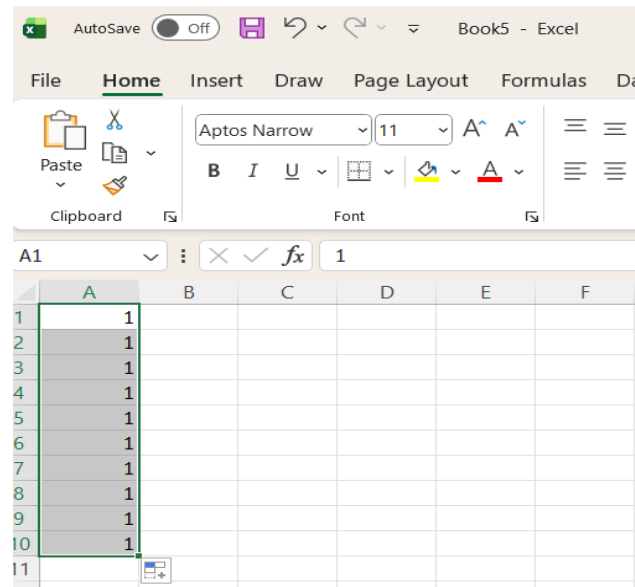


b. Select the Column and inset option

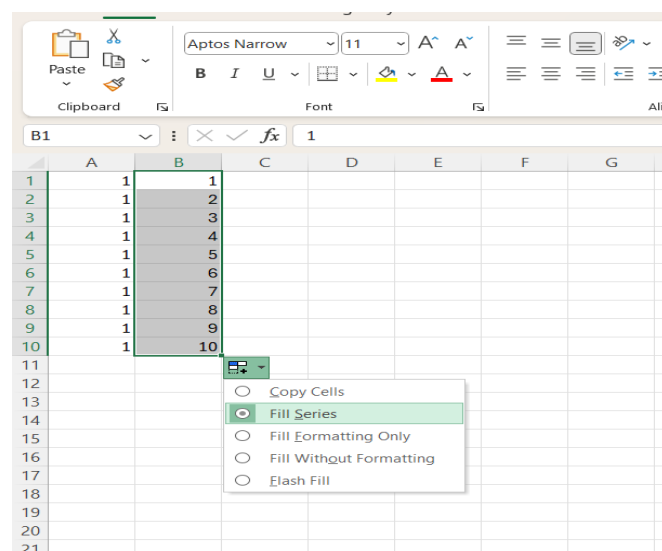


3. Drag and fill

a. Select the cell, a black plus sign appears as shown in the following image. This plus sign is the changed from of the fill handle, which appears as the cursor in placed on it. Drag the fill handle till cell you want. Hence, Excel has copied the value of the initially selected cell to a series of cells.



b. Fill in the series option This option works with numerical values or a mix of numerical and textual values. It fills the range with a series of values based on the sequence detected. Excel identifies a sequence when more than one data cell is selected. For instance, cells A1, A2, and A3 contain 1,2, and 3. Select these cells and drag the fill handle downwards. The range is filled with a series a value incrementing by 1.

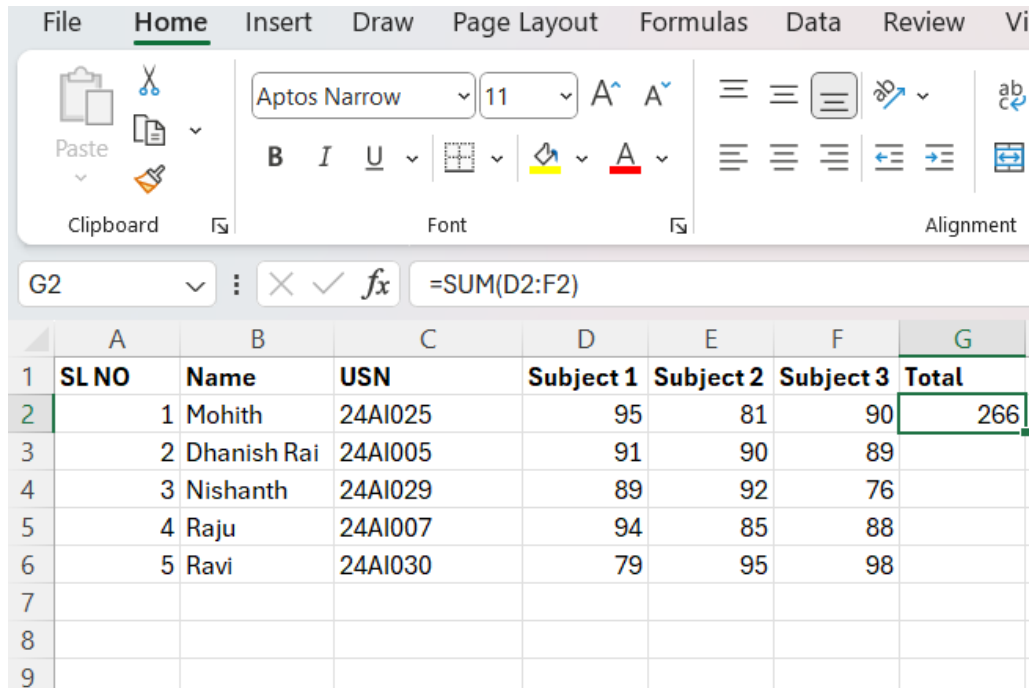


4. Use of aggregate functions:

To use the aggregate functions let us consider a student marks data.

a. Sum with Excel table: To calculate the total marks of all students.

=SUM(D2:F2)

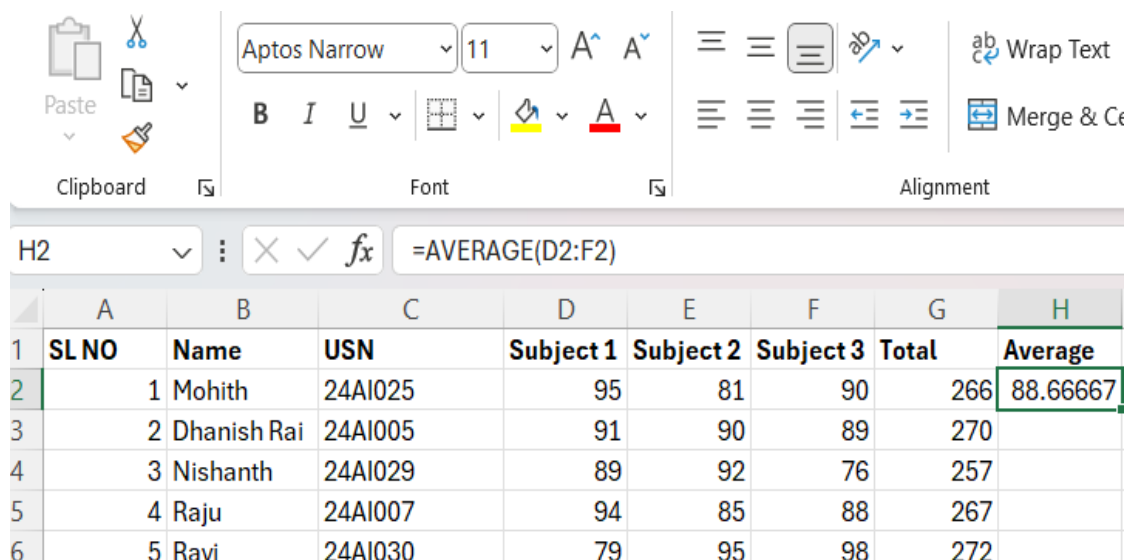


The screenshot shows the Microsoft Excel interface with the 'Home' tab selected. The formula bar displays '=SUM(D2:F2)'. Below the formula bar, a table is visible with the following data:

	A	B	C	D	E	F	G
1	SL NO	Name	USN	Subject 1	Subject 2	Subject 3	Total
2	1	Mohith	24AI025	95	81	90	266
3	2	Dhanish Rai	24AI005	91	90	89	
4	3	Nishanth	24AI029	89	92	76	
5	4	Raju	24AI007	94	85	88	
6	5	Ravi	24AI030	79	95	98	
7							
8							
9							

b. Average with Excel table: To find the average marks of the students

=AVERAGE(D2:F2)



The screenshot shows the Microsoft Excel interface with the 'Home' tab selected. The formula bar displays '=AVERAGE(D2:F2)'. Below the formula bar, a table is visible with the following data:

	A	B	C	D	E	F	G	H
1	SL NO	Name	USN	Subject 1	Subject 2	Subject 3	Total	Average
2	1	Mohith	24AI025	95	81	90	266	88.66667
3	2	Dhanish Rai	24AI005	91	90	89	270	
4	3	Nishanth	24AI029	89	92	76	257	
5	4	Raju	24AI007	94	85	88	267	
6	5	Ravi	24AI030	79	95	98	272	

=PRODUCT(D2:F2)

d. Minimum and Maximum in Excel table: To find the highest and lowest marks within the table.

=MAX(D2:F2)

Page | 5

=MIN(D2:F2)

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fx

=MIN(D2:F2)

B	C	D	E	F	G	H	I	J	K
ne	USN	Subject 1	Subject 2	Subject 3	Total	Average	Product	Maximum	Minimum
ith	24AI025	95	81	90	266	88.66667	692550	95	81
nish Rai	24AI005	91	90	89	270	90	728910	91	
nanth	24AI029	89	92	76	257	85.66667	622288	92	
i	24AI007	94	85	88	267	89	703120	94	
i	24AI030	79	95	98	272	90.66667	735490	98	

e. Count in Excel table: Counts the number of cells in a range that contains numbers

=COUNT(A1:K6)

Clipboard Font Alignment Number

D10 : X ✓ fx =COUNT(A1:K6)

	A	B	C	D	E	F	G	H	I	J	K
1	SLNO	Name	USN	Subject 1	Subject 2	Subject 3	Total	Average	Product	Maximum	Minimum
2	1	Mohith	24AI025	95	81	90	266	88.66667	692550	95	81
3	2	Dhanish Rai	24AI005	91	90	89	270	90	728910	91	89
4	3	Nishanth	24AI029	89	92	76	257	85.66667	622288	92	76
5	4	Raju	24AI007	94	85	88	267	89	703120	94	85
6	5	Ravi	24AI030	79	95	98	272	90.66667	735490	98	79
7											
8											
9				COUNT							
10				45							
11											
12											

f. Counta in Excel table: Counts the number of cells in range that are nonempty

=COUNTA(E2:E6)

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E10







=COUNTA(E2:E6)

	A	B	C	D	E	F	G	H	I	J	K
1	SLNO	Name	USN	Subject 1	Subject 2	Subject 3	Total	Average	Product	Maximum	Minimum
2	1	Mohith	24AI025	95	81	90	266	88.66667	692550	95	81
3	2	Dhanish Rai	24AI005	91	90	89	270	90	728910	91	89
4	3	Nishanth	24AI029	89		76	165	82.5	6764	89	76
5	4	Raju	24AI007	94	85	88	267	89	703120	94	85
6	5	Ravi	24AI030	79	95	98	272	90.66667	735490	98	79
7											
8											
9											
10				COUNT	COUNTA						
11				44	4						
12											

j. Countblank in Excel table: Counts number of empty cells in the specified range

=COUNT(E2:E6)

FileHomeInsertDrawPage LayoutFormulasDataReviewViewHelpAcrobat

Paste

Clipboard

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11

Wrap Text

Merge & Center

General

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F10

=COUNTBLANK(E2:E6)

SL NO	Name	USN	Subject 1	Subject 2	Subject 3	Total	Average	Product	Maximum	Minimum
1	Mohith	24AI025	95	81	90	266	88.66667	692550	95	81
2	Dhanish Rai	24AI005	91	90	89	270	90	728910	91	89
3	Nishanth	24AI029	89		76	165	82.5	6764	89	76
4	Raju	24AI007	94	85	88	267	89	703120	94	85
5	Ravi	24AI030	79	95	98	272	90.66667	735490	98	79
										</

EXPERIMENT 2: WORKING WITH DATA

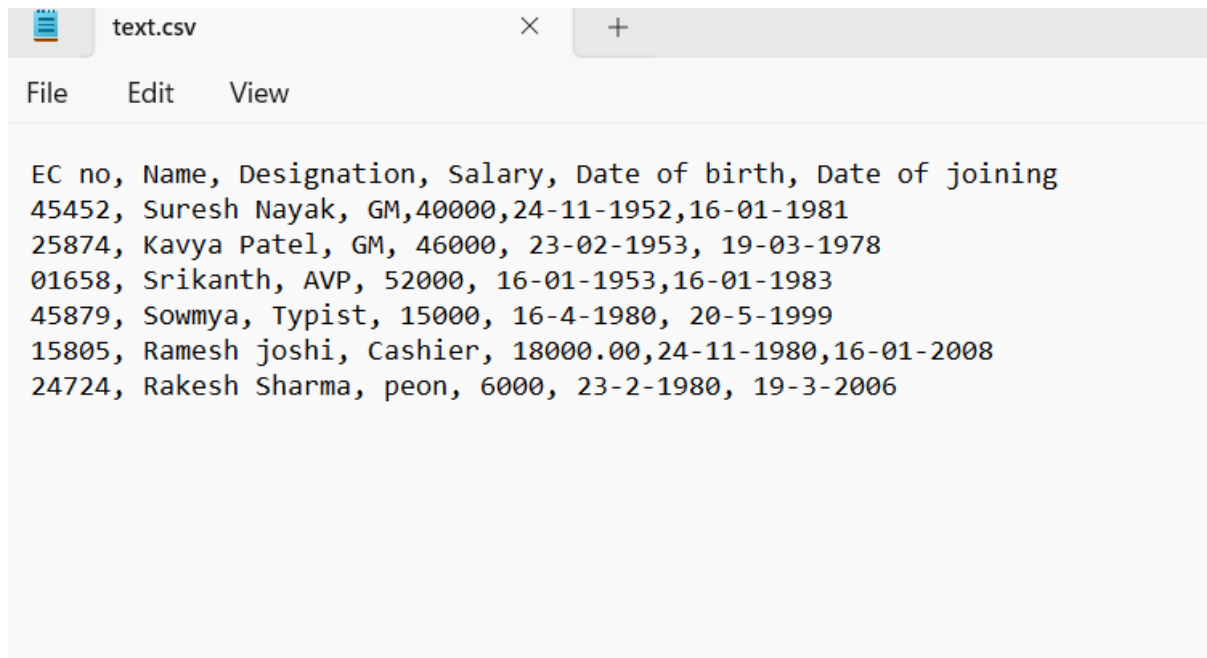
1. Importing Data from a Text File

If we have a data in .txt, or .csv files we can import the data from those files treating them as a text files. Importing text or csv files into Excel creates a dynamic connection that allows you automate data cleaning and bypass standard row limits. It ensures data integrity by preventing Excel from auto-formatting values like dates or phone numbers.

The steps for the importing data from the csv file are:

Step 1- Open the notepad.

Step 2- Write the data in the notepad that you want to import to the excel as shown below.



```
EC no, Name, Designation, Salary, Date of birth, Date of joining
45452, Suresh Nayak, GM,40000,24-11-1952,16-01-1981
25874, Kavya Patel, GM, 46000, 23-02-1953, 19-03-1978
01658, Srikanth, AVP, 52000, 16-01-1953,16-01-1983
45879, Sowmya, Typist, 15000, 16-4-1980, 20-5-1999
15805, Ramesh joshi, Cashier, 18000.00,24-11-1980,16-01-2008
24724, Rakesh Sharma, peon, 6000, 23-2-1980, 19-3-2006
```

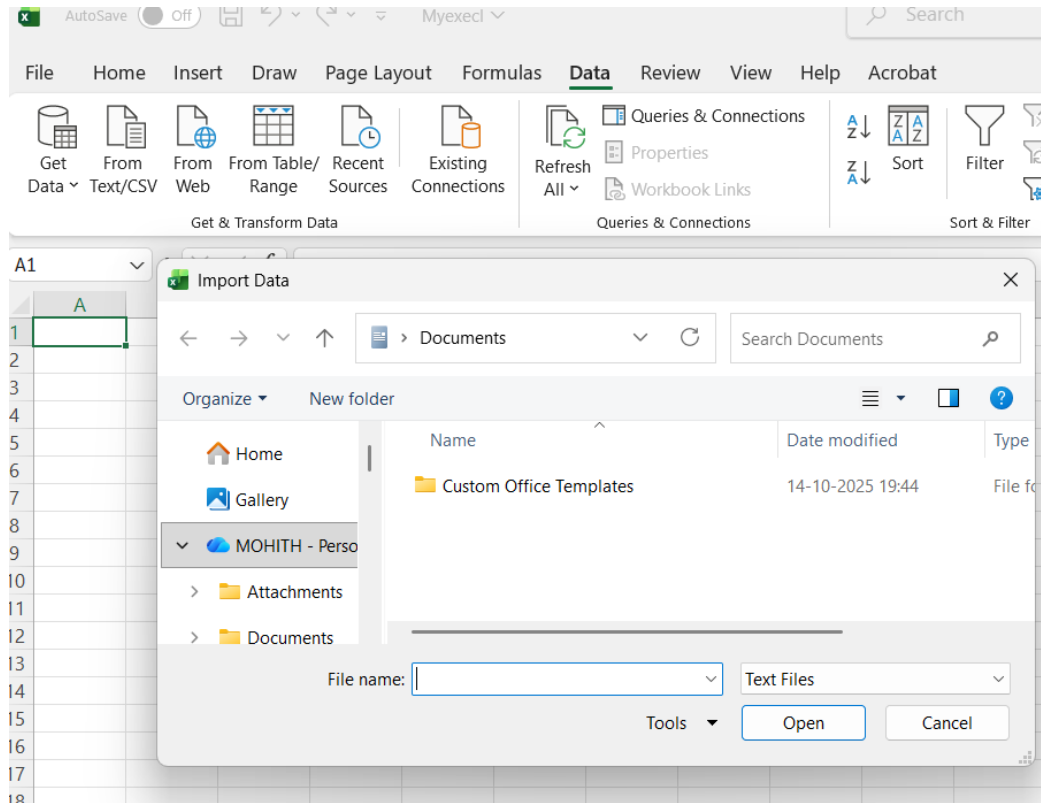
Rules to follow while writing the csv file:

- i) One record per Line: Each row of data must stay on its own line, separated by line break and each data separated by comma.
- ii) Consistent Column Count: Every row must have the exact same number of columns. if a row has missing data, leave it empty between the commas (e.g., Value1, , Value2) rather than deleting the comma.
- iii) Header row: The first row should contain unique, descriptive column name. Avoid using special characters or spaces in headers
- iv) Save the file as .csv extension anywhere in your system.

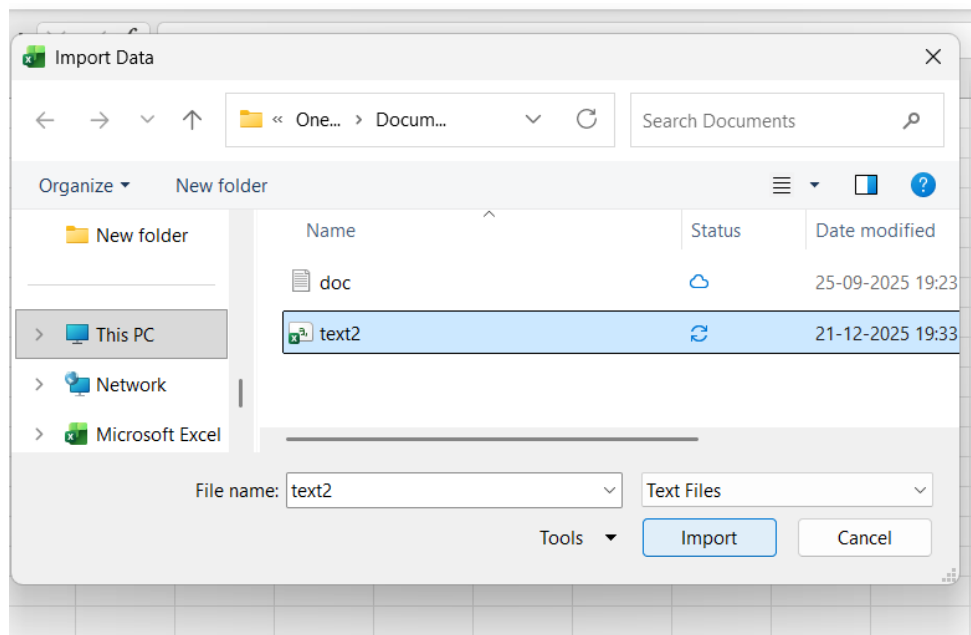
Step3-Open Excel and select a new sheet.

Step4-Click the DATA tab on the Ribbon.

Step5-Click **from Text/csv**. The **Import Text File** dialog box appears.



Step6-Click on the file that you saved earlier that contains the data that need to be imported.
and click import.



Step7-A dialogue box appears click on **Load**.

text2.csv

File Origin: 1252: Western European (Windows) | Delimiter: Comma | Data Type Detection: Based on first 200 rows

EC no	Name	Designation	Salary	Date of birth	Date of joining
45452	Suresh Nayak	GM	40000	24-11-1952	16-01-1981
25874	Kavya Patel	GM	46000	23-02-1953	19-03-1978
1658	Srikanth	AVP	52000	16-01-1953	16-01-1983
45879	Sowmya	Typist	15000	16-04-1980	20-05-1999
15805	Ramesh joshi	Cashier	18000	24-11-1980	16-01-2008
24724	Rakesh Sharma	peon	6000	23-02-1980	19-03-2006

Buttons: Load, Transform Data, Cancel

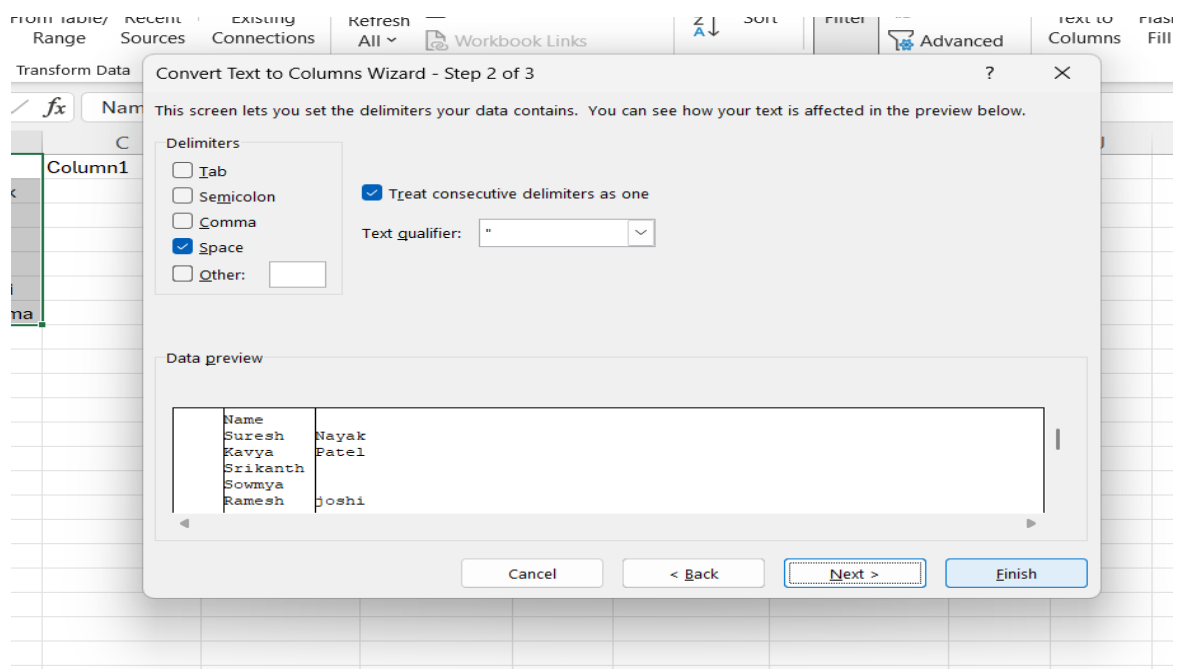
All the data from the text file now has been imported into the Excel

File Home Insert Draw Page Layout Formulas Data Review View Help ACR

Clipboard: Paste, Font: Aptos Narrow, 11, Bold, Italic, Underline, Font Color, Background Color, Alignment: Wrap Text, Merge & Center

Formula Bar: G7

EC no	Name	Designation	Salary	Date of birth	Date of joining
45452	Suresh Nayak	GM	40000	24-11-1952	16-01-1981
25874	Kavya Patel	GM	46000	23-02-1953	19-03-1978
1658	Srikanth	AVP	52000	16-01-1953	16-01-1983
45879	Sowmya	Typist	15000	16-04-1980	20-05-1999
15805	Ramesh joshi	Cashier	18000	24-11-1980	16-01-2008
24724	Rakesh Sharma	peon	6000	23-02-1980	19-03-2006



Get & Transform Data				Queries & Connections			Sort
D10							
	A	B	C	D	E	F	G
1	EC no	Name	Surname	Designation	Salary	Date of birth	Date of joining
2	45452	Suresh	Nayak	GM	40000	24-11-1952	16-01-1981
3	25874	Kavya	Patel	GM	46000	23-02-1953	19-03-1978
4	1658	Srikanth		AVP	52000	16-01-1953	16-01-1983
5	45879	Sowmya		Typist	15000	16-04-1980	20-05-1999
6	15805	Ramesh	joshi	Cashier	18000	24-11-1980	16-01-2008
7	24724	Rakesh	Sharma	peon	6000	23-02-1980	19-03-2006
8							
9							
10							
11							
12							
13							
14							
15							

b) Sorting:

Arranging the data in the particular order (ascending order or descending order).

Step 1: Select required data and go to data tab and click on sort.

25874	Kavya	Patel
1658	Srikanth	
45879	Sowmya	
15805	Ramesh	joshi
24724	Rakesh	Sharma

Copy

Paste Options:

Paste Special...

 Refresh

Insert >

Delete >

Select >

Clear Contents

 Quick Analysis

Sort >

Filter >

Table >

 Get Data from Table/Range...

New Comment

New Note

23-02-1953	19-03-1978
16-01-1953	16-01-1983
16-04-1980	20-05-1999
24-11-1980	16-01-2008
23-02-1980	19-03-2006

Sort A to Z

Sort Z to A

Put Selected Cell Color On Top

Put Selected Font Color On Top

Put Selected Formating Icon On Top

Custom Sort...

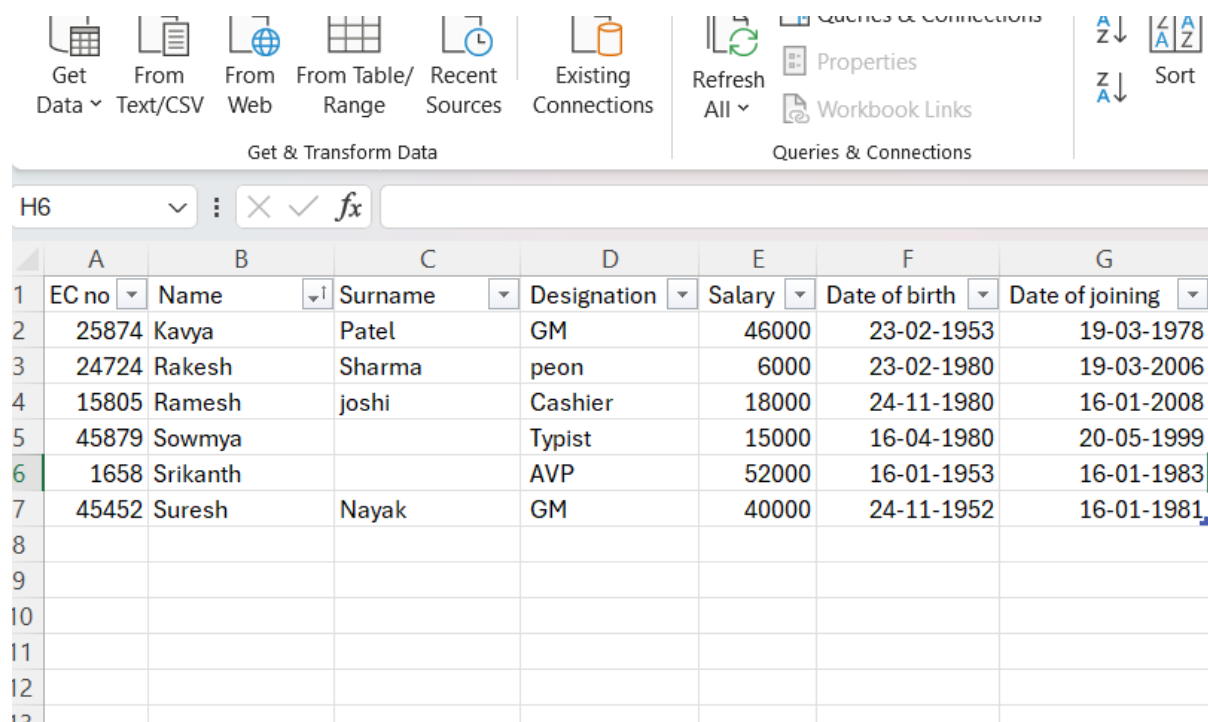
After sorting data, data looks like this

Get & Transform Data				Queries & Connections			Sc
B1							
A	B	C	D	E	F	G	
EC no	Name	Surname	Designation	Salary	Date of birth	Date of joining	
25874	Kavya	Patel	GM	46000	23-02-1953	19-03-1978	
24724	Rakesh	Sharma	peon	6000	23-02-1980	19-03-2006	
15805	Ramesh	joshi	Cashier	18000	24-11-1980	16-01-2008	
45879	Sowmya		Typist	15000	16-04-1980	20-05-1999	
1658	Srikanth		AVP	52000	16-01-1953	16-01-1983	
45452	Suresh	Nayak	GM	40000	24-11-1952	16-01-1981	

c)Filtering:

Selecting specific subsets of data based on criteria.

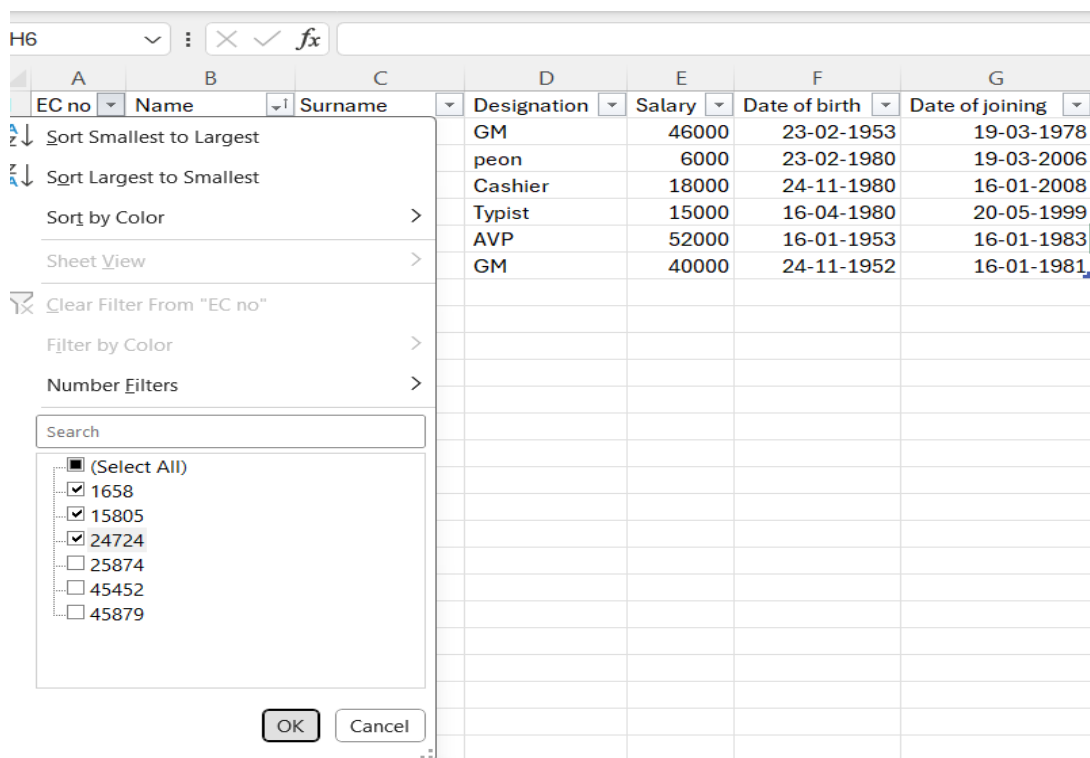
Step 1: Select the data range including the column headings and go to data tab and click on filter.



	A	B	C	D	E	F	G
1	EC no	Name	Surname	Designation	Salary	Date of birth	Date of joining
2	25874	Kavya	Patel	GM	46000	23-02-1953	19-03-1978
3	24724	Rakesh	Sharma	peon	6000	23-02-1980	19-03-2006
4	15805	Ramesh	joshi	Cashier	18000	24-11-1980	16-01-2008
5	45879	Sowmya		Typist	15000	16-04-1980	20-05-1999
6	1658	Srikanth		AVP	52000	16-01-1953	16-01-1983
7	45452	Suresh	Nayak	GM	40000	24-11-1952	16-01-1981
8							
9							
10							
11							
12							
13							

Step 2: Click the drop-down arrow in the required column header and select the values to be displayed.

Step 3: Click “OK” to apply the filter and view the selected records only.



	A	B	C	D	E	F	G
1	EC no	Name	Surname	Designation	Salary	Date of birth	Date of joining
2	25874	Kavya	Patel	GM	46000	23-02-1953	19-03-1978
3	24724	Rakesh	Sharma	peon	6000	23-02-1980	19-03-2006
4	15805	Ramesh	joshi	Cashier	18000	24-11-1980	16-01-2008
5	45879	Sowmya		Typist	15000	16-04-1980	20-05-1999
6	1658	Srikanth		AVP	52000	16-01-1953	16-01-1983
7	45452	Suresh	Nayak	GM	40000	24-11-1952	16-01-1981
8							
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10							
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12							
13							

AutoSave Off Myexcel Search

File Home Insert Draw Page Layout Formulas **Data** Review View Help Acrobat

Get Data From Text/CSV From Web From Table/Range Recent Sources Existing Connections Refresh All Queries & Connections Properties Workbook Links Sort Filter

Get & Transform Data Queries & Connections Sort & Filter

H6

	A	B	C	D	E	F	G	H
1	EC no	Name	Surname	Designation	Salary	Date of birth	Date of joining	
3	24724	Rakesh	Sharma	peon	6000	23-02-1980	19-03-2006	
4	15805	Ramesh	joshi	Cashier	18000	24-11-1980	16-01-2008	
6	1658	Srikanth		AVP	52000	16-01-1953	16-01-1983	
8								
9								
10								
11								
12								
13								

EXPERIMENT 3: DATA REPORTING

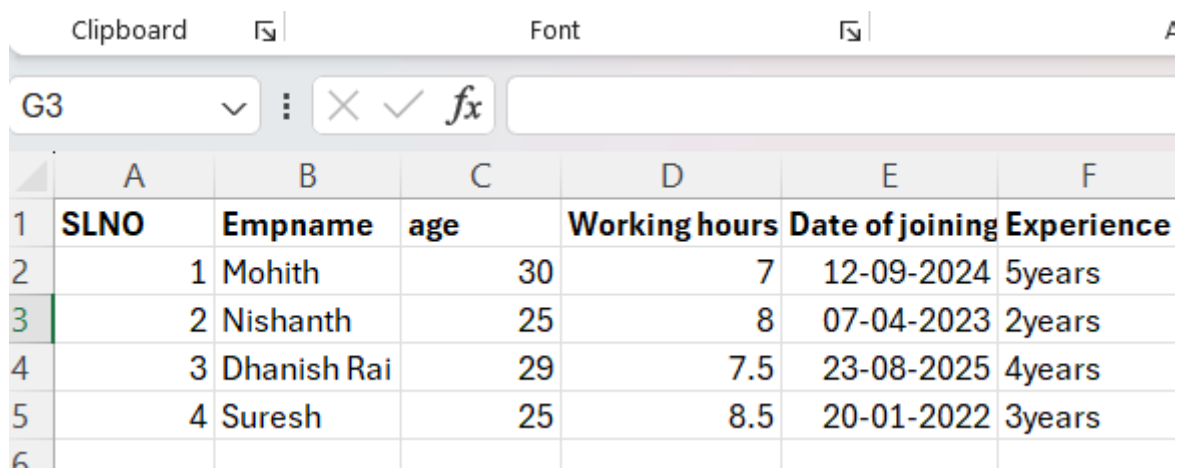
1. Data Validation

Data Validation is a feature that allows you to control and restrict the type of data that can be entered into a cell. This ensures data accuracy, consistency, and reliability, which is crucial in data analytics.

Data validation allows you to define rules or conditions for the data that can be entered in a cell. These rules can include specifying acceptable values, numbers formats, text lengths, dates, or custom formulas. Data Validation helps prevent errors and inconsistencies in your dataset.

a. Whole number:

Let's consider employee details data

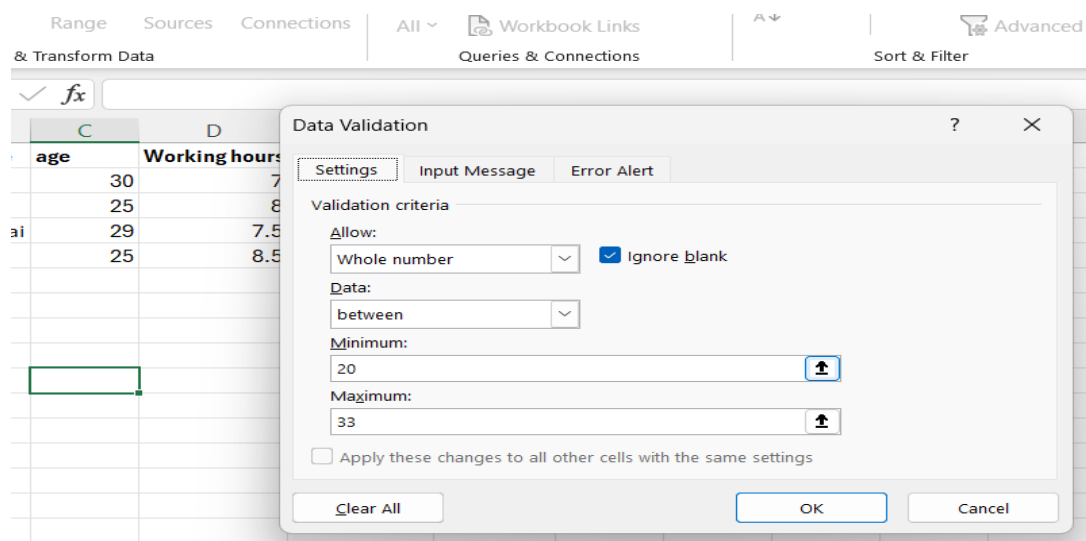


	A	B	C	D	E	F
	SLNO	Empname	age	Working hours	Date of joining	Experience
2	1	Mohith	30	7	12-09-2024	5years
3	2	Nishanth	25	8	07-04-2023	2years
4	3	Dhanish Rai	29	7.5	23-08-2025	4years
5	4	Suresh	25	8.5	20-01-2022	3years

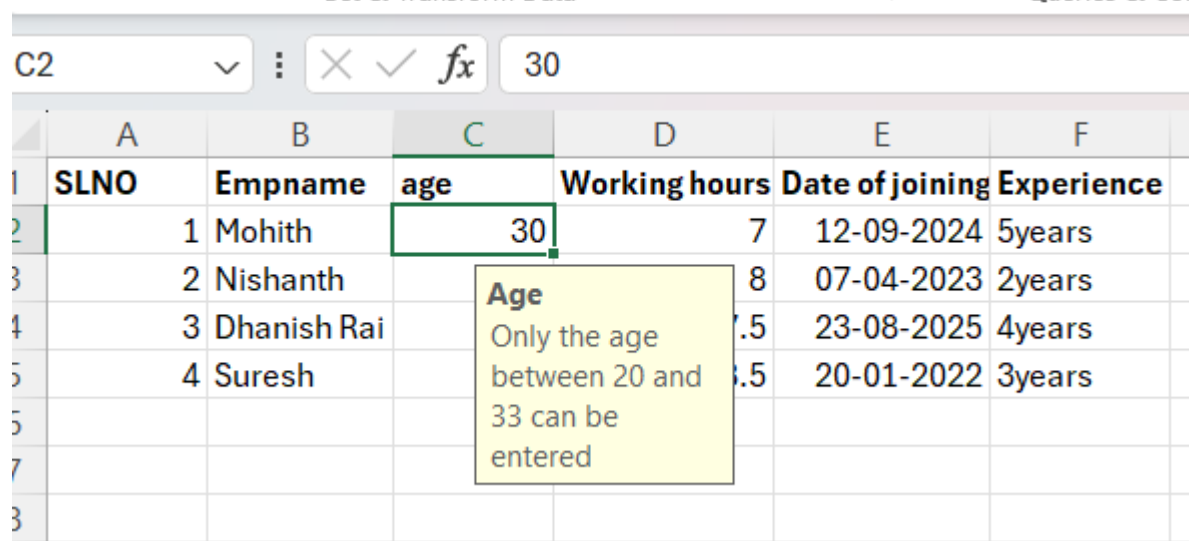
Let's apply data validation to "age" column to ensure that only age between 18 and 60 are accepted.

Step1: Select the cells in the "age" column where you want to apply data validation.

Step2: Go to "data" tab then click on "data validation". Allow whole numbers and set the required conditions. Click OK to apply data validation to the selected cells.

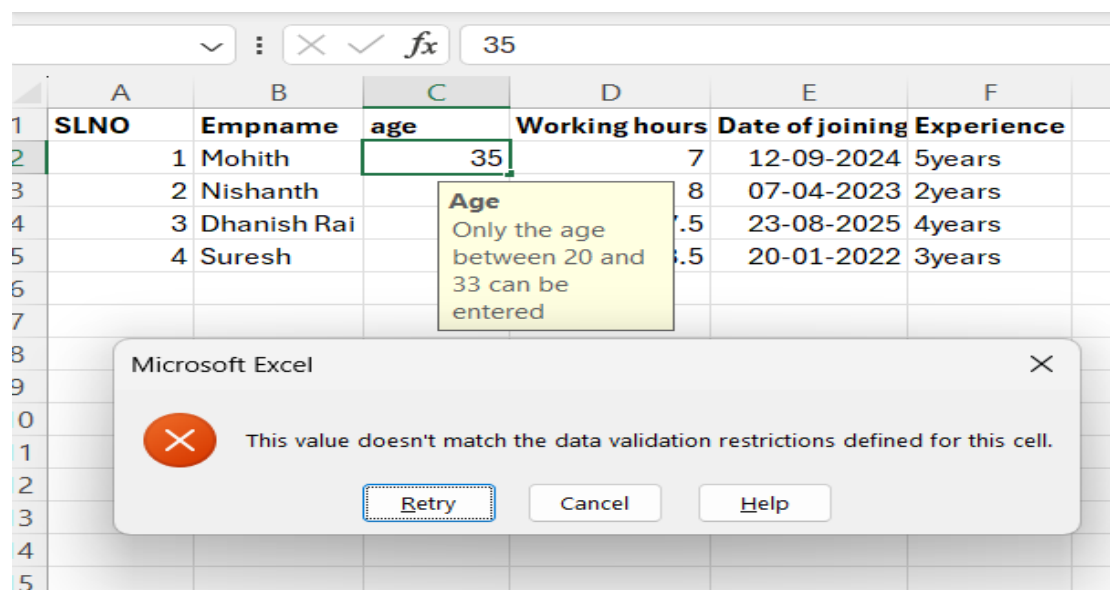


Step3: In the data validation dialog box, click on the “input message” tab and enter a title and message to guide the user while entering data.



	A	B	C	D	E	F
1	SLNO	Empname	age	Working hours	Date of joining	Experience
2	1	Mohith	30	7	12-09-2024	5years
3	2	Nishanth		8	07-04-2023	2years
4	3	Dhanish Rai		5.5	23-08-2025	4years
5	4	Suresh		5.5	20-01-2022	3years

Step4: Click on the “error alert”, enter the title and error message that appears when invalid data is entered.

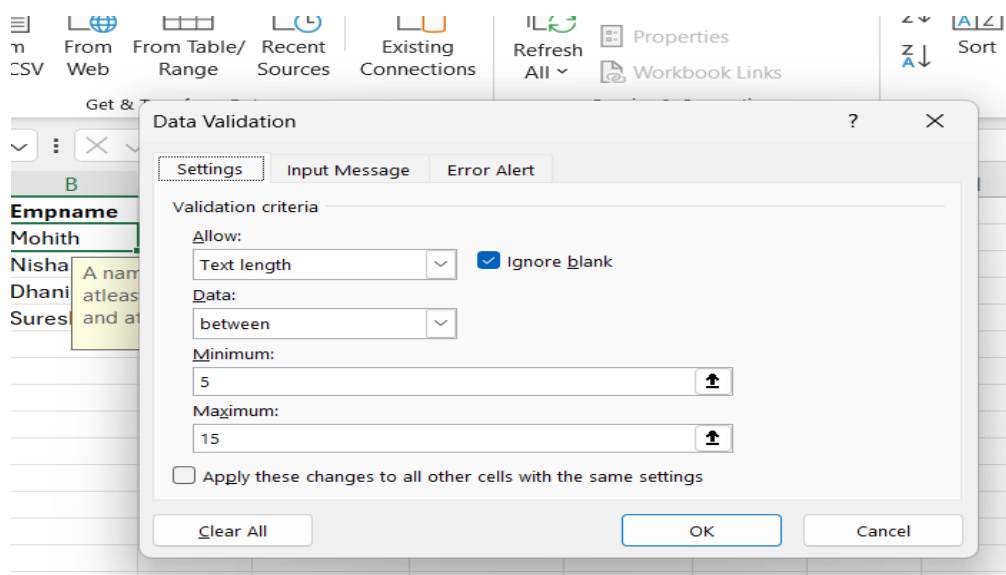


	A	B	C	D	E	F
1	SLNO	Empname	age	Working hours	Date of joining	Experience
2	1	Mohith	35	7	12-09-2024	5years
3	2	Nishanth		8	07-04-2023	2years
4	3	Dhanish Rai		5.5	23-08-2025	4years
5	4	Suresh		5.5	20-01-2022	3years

Now whenever the data is entered in age column a message occurs. If the age is entered above 33 error message occurs.

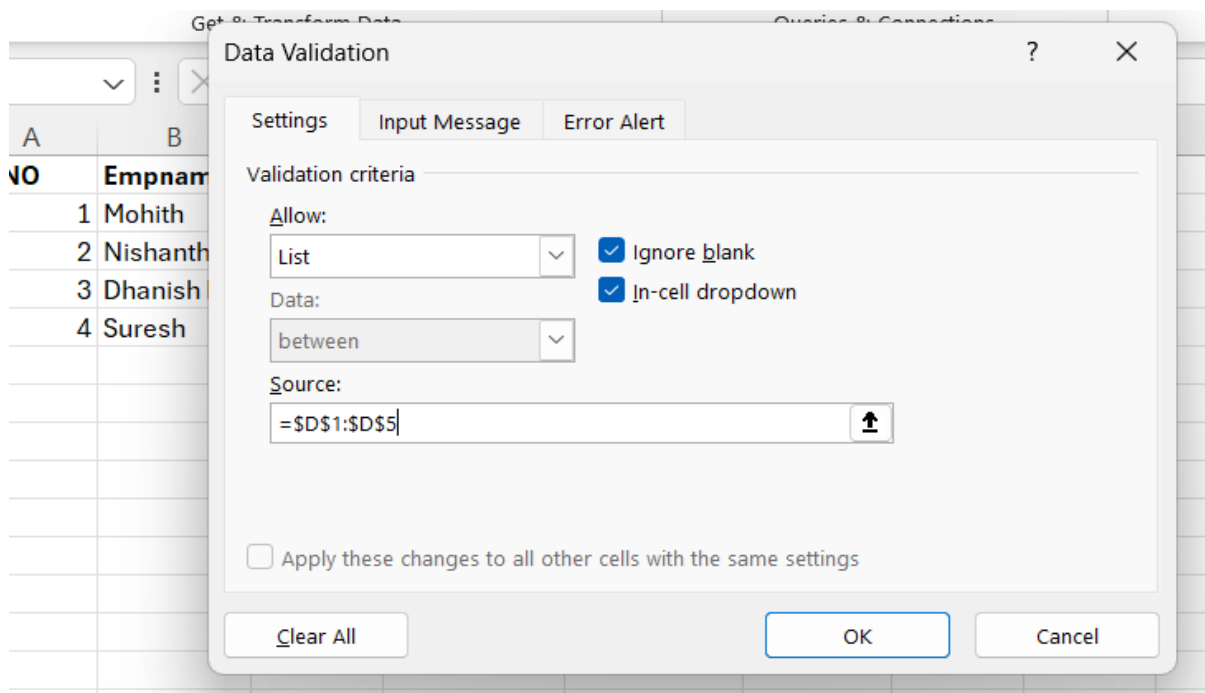
b. Text length:

Now let’s do the data validation for the column “Empname”, by keeping the text length



c. List: When list is selected, a drop-down list is created from which the user can choose predefined values.

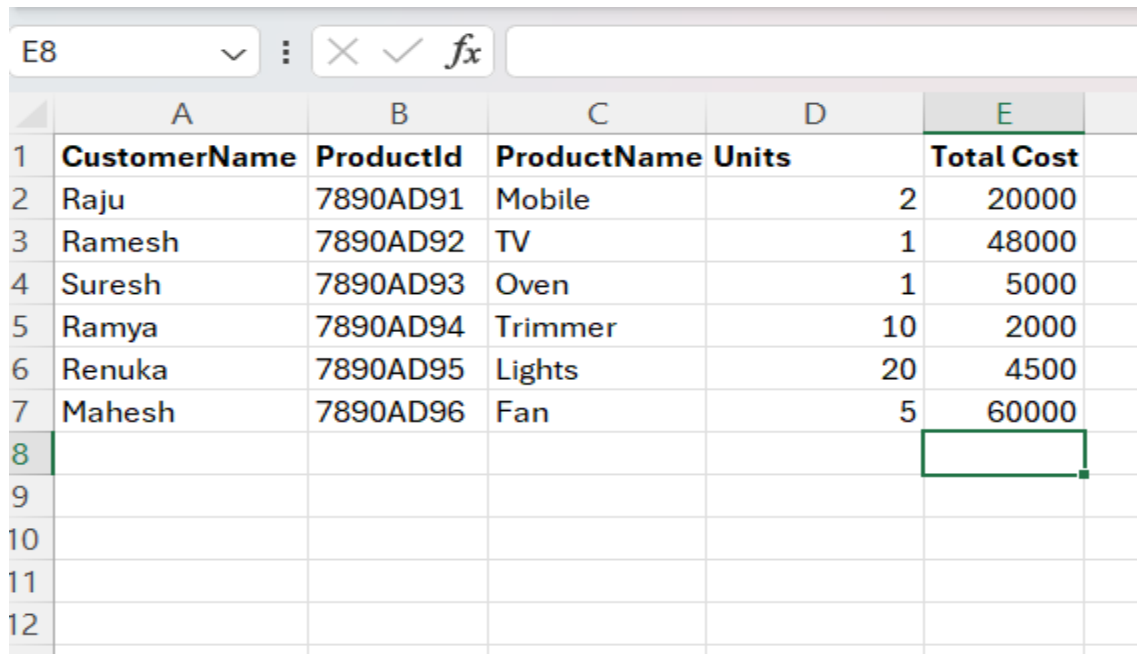
Let's create list for the "Working hours",



Also add the input Message and Error Alert to the list and whenever the wrong data is entered to the list the error message is shown.

2.Pivot table:

Pivot table is functionality which helps us organize and analyse data.

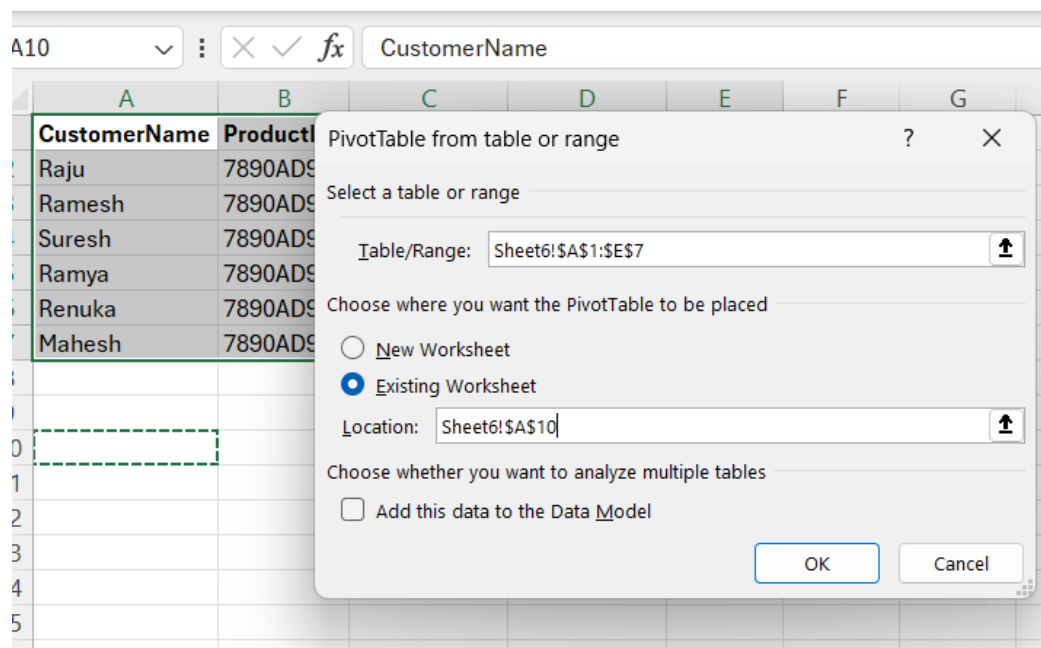


	A	B	C	D	E
1	CustomerName	ProductId	ProductName	Units	Total Cost
2	Raju	7890AD91	Mobile	2	20000
3	Ramesh	7890AD92	TV	1	48000
4	Suresh	7890AD93	Oven	1	5000
5	Ramya	7890AD94	Trimmer	10	2000
6	Renuka	7890AD95	Lights	20	4500
7	Mahesh	7890AD96	Fan	5	60000
8					
9					
10					
11					
12					

Step 1: Select the entire data range (including headers) and go to insert tab then select “pivot table”.

Step 2: A create PivotTable dialog box will appear, choose where you want to place the pivot table click on “existing worksheet”.

Step 3: Select a location (cell) to place the pivot table. Click on OK.



Step 4: Create a pivot table. In the Pivot Table Fields pane,

- Drag the “Customername and Productid” field to the “rows” area.
- Drag the “Productid” field to the “columns” area.
- Drag the “total cost” field to the “values” area and ensure it is set to **Sum of Sales**.
- The pivot table will display Customername and Productid sales along with Grand Total automatically. Check whether all fields (productid, total) are placed correctly. Click on the drop-down arrow in row labels or column labels to filter specific name or productid. Use sort option to arrange sales values in ascending or descending order.

3.Pivot Chart:

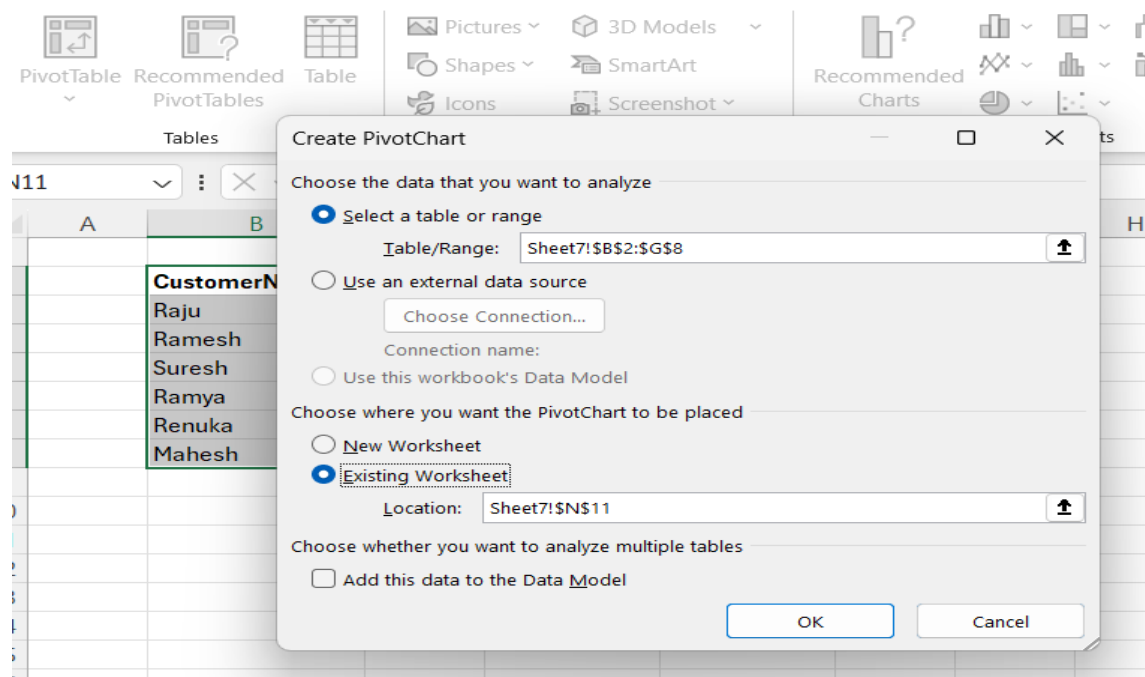
A Pivot chart is a graphical representation that is directly linked to a pivot table. Pivot charts are used to visualize and present summarized data from a pivot table in a more visually appealing and understandable way.

To create pivot chart for the same above table but let’s add a region column follow the following steps:

Step1: Select the entire data range (including headers) and go to insert tab then select “pivot chart”.

Step 2: A create PivotChart dialog box will appear, choose where you want to place the pivot table click on “existing worksheet”.

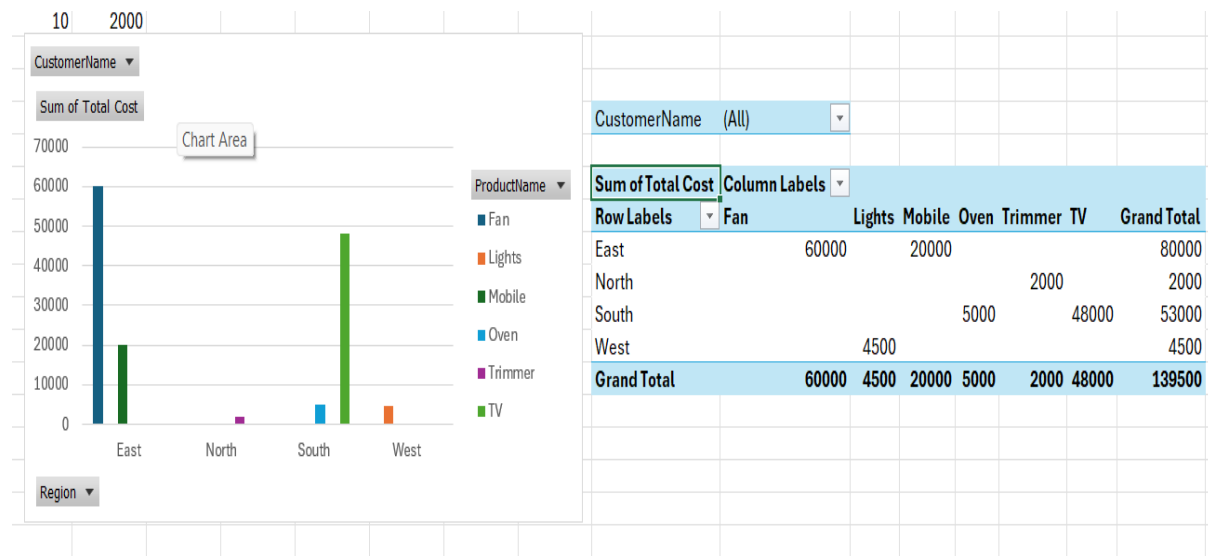
Step 3: Select a location (cell) to place the pivot chart. Click on OK.



Step4: Create a pivot chart. In the PivotChart Fields pane,

- Drag the “Region” field to the “rows” area.
- Drag the “ProductName” field to the “columns” area.
- Drag the “Total cost” field to the “values” area and ensure it is set to Sum of Sales.
- Drag the “CustomerName” field to the “filters” area.

Click on the drop-down arrow in row labels or column labels to filter specific regions or products.

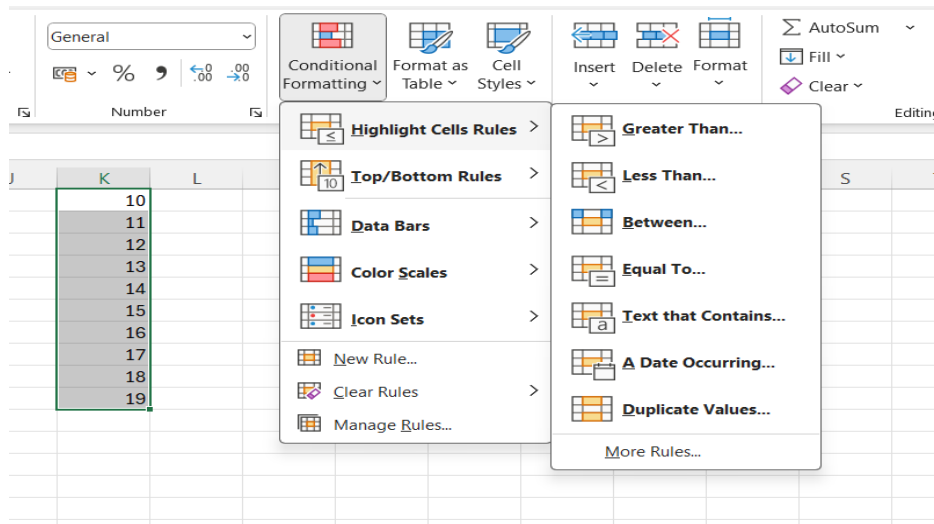


EXPERIMENT NO.4: Data Analysis Process

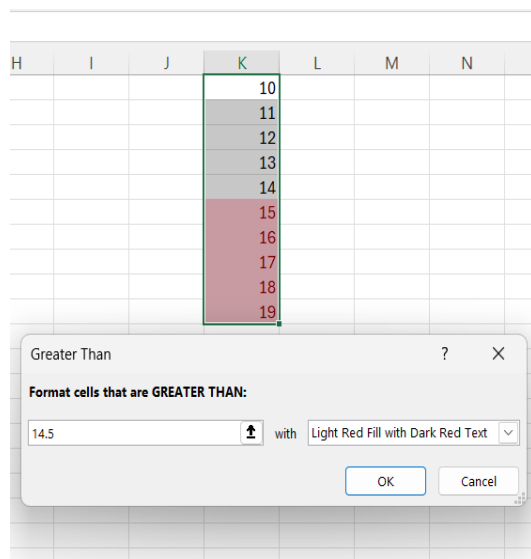
1.Conditional formatting:

Conditional formatting is a feature that helps you highlight data without manually formatting it. Using conditional formatting you can change cell colour, add data bars, add colour scales, add icons etc.

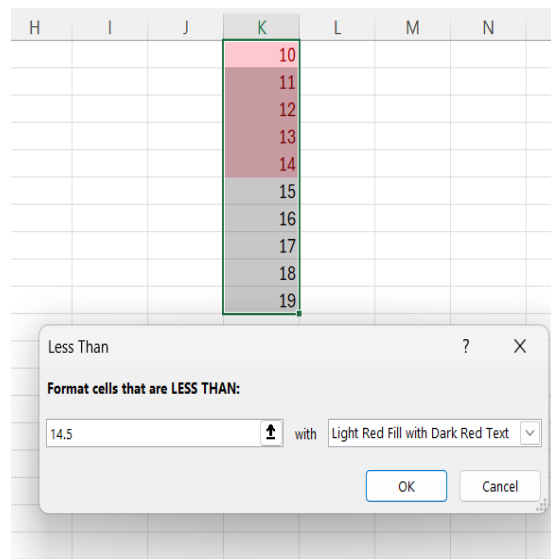
Steps: Select cells>Home>Conditional formatting (Highlight cells that have certain conditions)>Highlight cells with rules>



a. Greater than

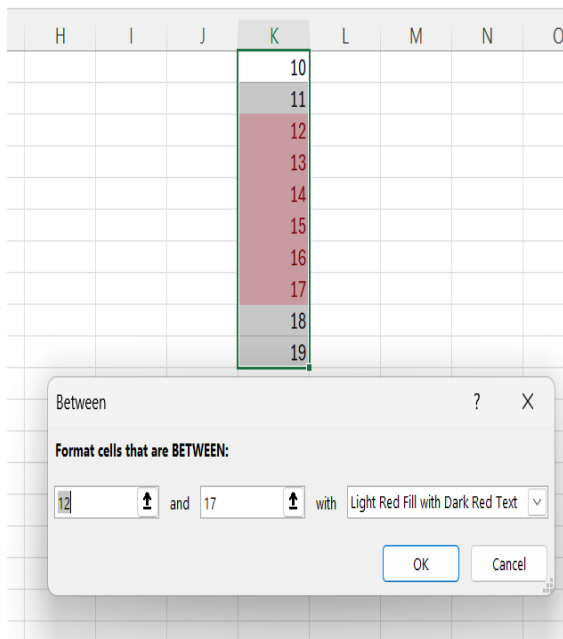


b. Lesser than

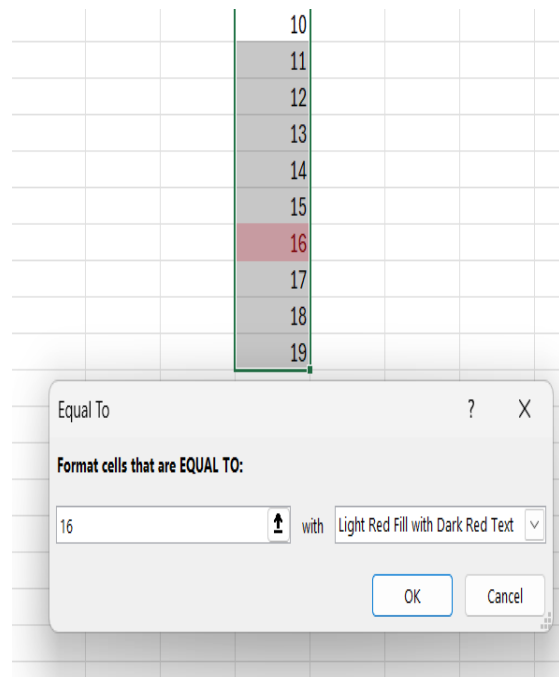


The highlighted cells in both greater than and lesser then shows the conditions given.

c. Between

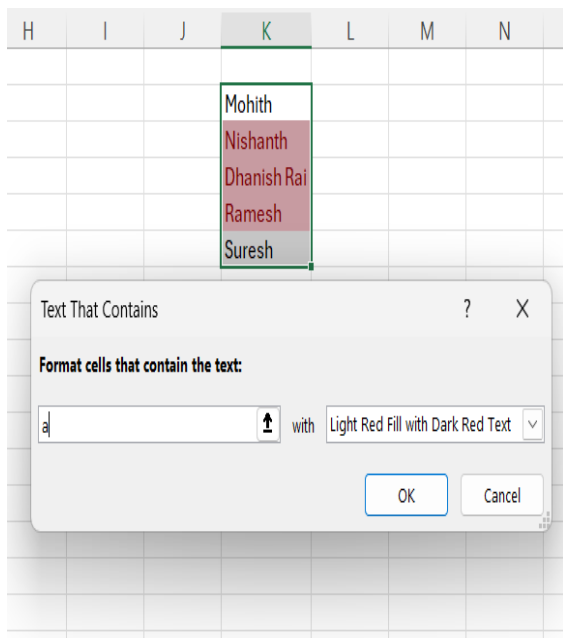


d. Equal to

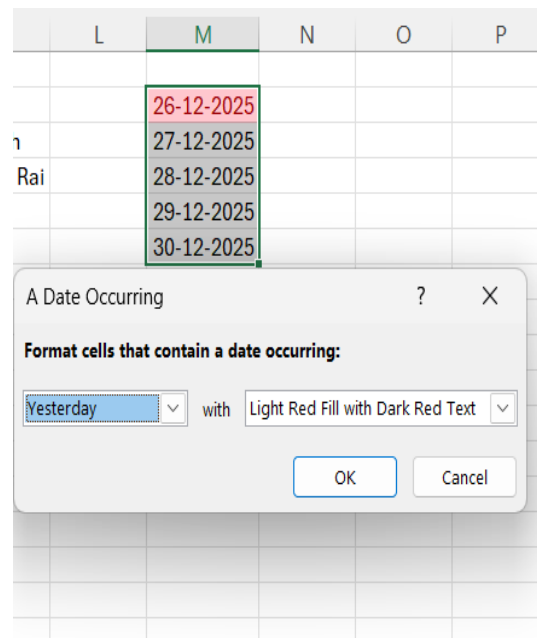


The “Between” function highlights the number specified in the range and “Equal to” highlights exact number given.

e. text that contains

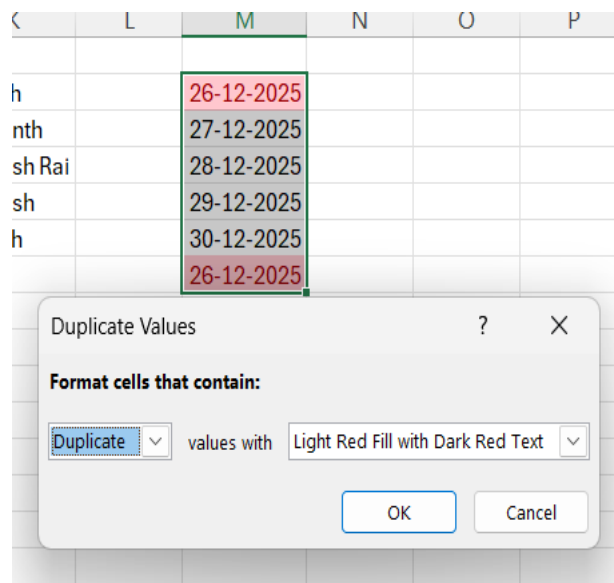


f. Date occurring



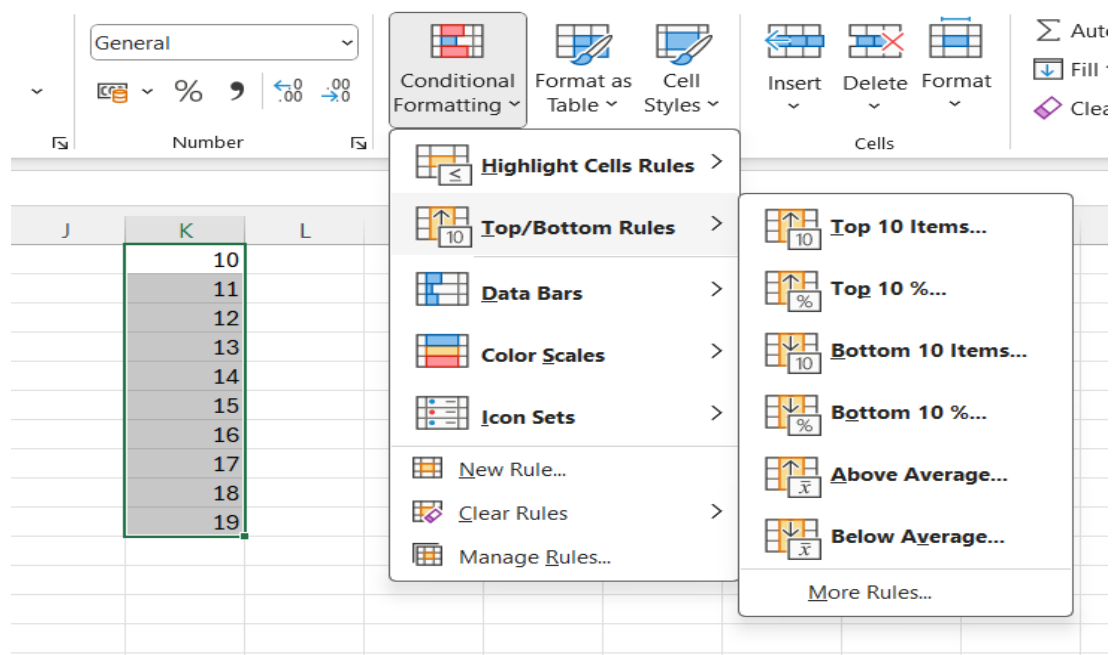
The “text that contains” highlights cells that contain the text given and “Date occurring” highlights the date specified.

g. Duplicate values:



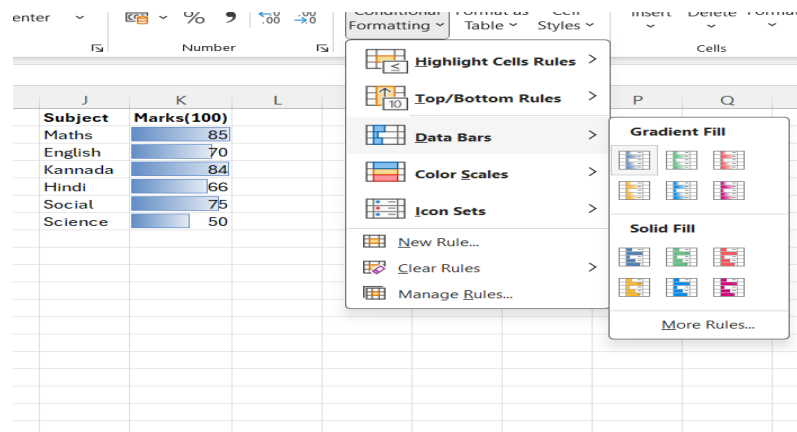
The “Duplicate values” highlights cells that contain the duplicate values in selected cells.

Top/Bottom cells



Data bars:

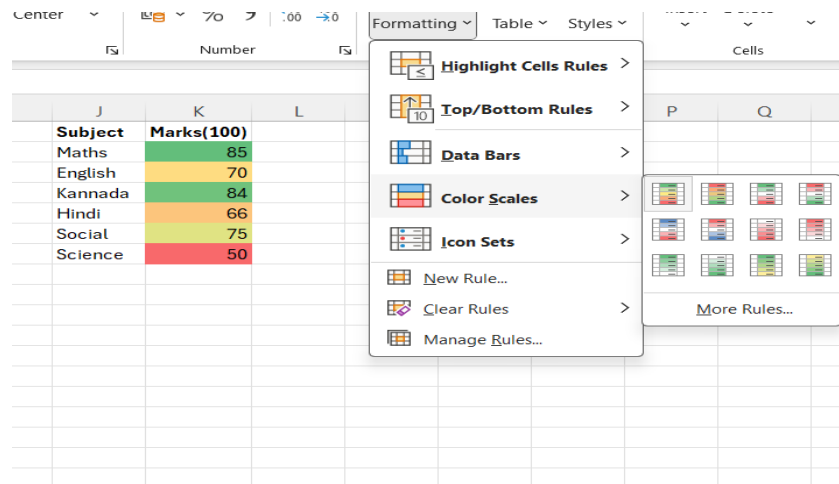
Select cells>Home>Conditional formatting> data bars



Colour

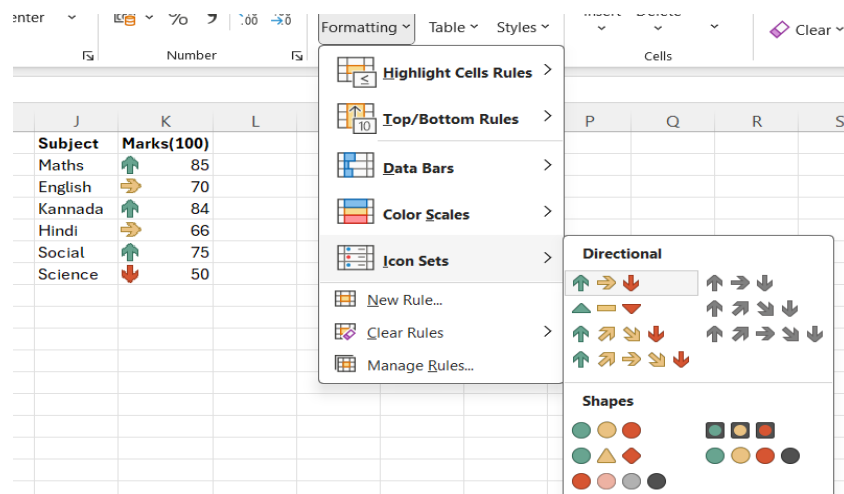
scales:

Select cells>Home>Conditional formatting> Colour scales



Icon Sets:

Select cells>Home>Conditional formatting> Icon sets



2.What if analysis:

In the new sheet add this data and calculate the using the formula given.

Price

Qty

Total Revenue=Pri*Qty

Transport Cost =TR*10%

Item Cost=20*Qty

Total Cost=Transport Cost +IC

Profit=TR-Total Cost

B	C	D	E	F	G
		Price	32		
		Qty	100		
		Total Revenue	3200		
		TransportCost	320		
		Item Cost	2000		
		TotalCost	2320		
		Profit	880		

Step 1: Select Qty cell (Changing cell) and Go to Data tab click on **What-if Analysis**, Select **Scenario Manager**.

The screenshot shows the Excel Data Tools ribbon. The 'What-If Analysis' dropdown menu is open, displaying three options: 'Scenario Manager...', 'Goal Seek...', and 'Data Table...'. The 'Scenario Manager...' option is highlighted. Below the ribbon, a portion of the spreadsheet is visible, showing the same data as the table above, with the 'Qty' cell (C2) selected.

Step 2: Click on Add to create a new scenario. Enter the scenario name.

Price	32
Qty	100
Total Revenue	3200
TransportCost	320
Item Cost	2000
TotalCost	2320
Profit	880

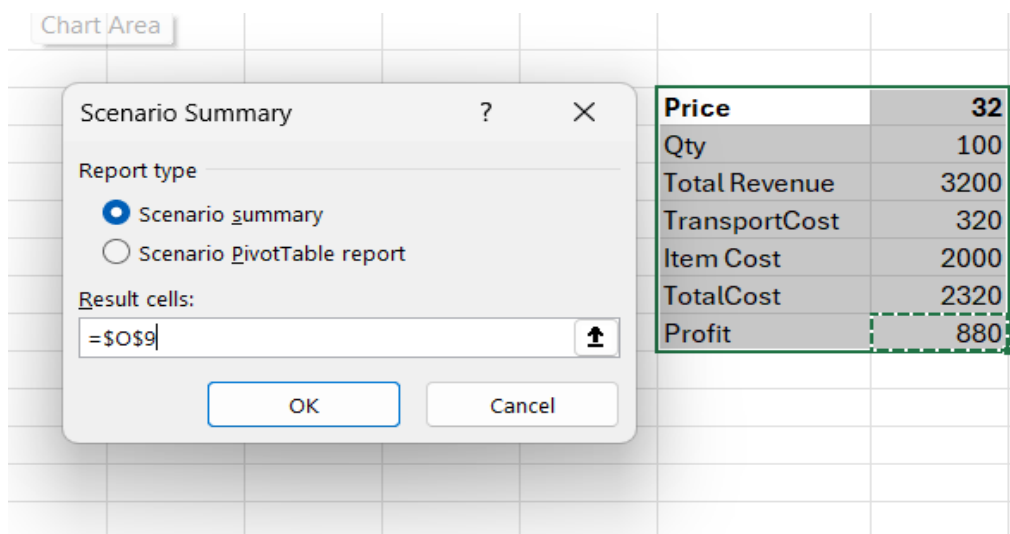
Step 3: Enter values for each of the changing cells. Click OK to save the scenario.

Price	32
Qty	100
Total Revenue	3200
TransportCost	320
Item Cost	2000
TotalCost	2320
Profit	880

Step 4: Repeat the steps to add more scenarios. Select any Scenario and click Summary.

Price	32
Qty	100
Total Revenue	3200
TransportCost	320
Item Cost	2000
TotalCost	2320
Profit	880

Step 5: Select cell profit for the result of cells. Click OK.



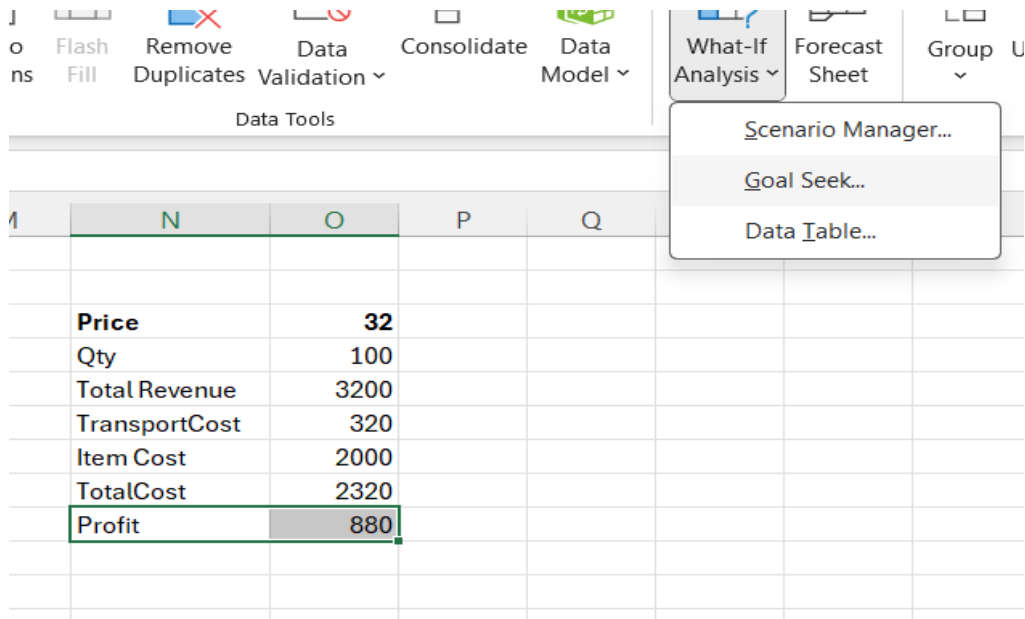
A Scenario Summary report is generated showing:

- Changing cell values.
- Corresponding profit values for each scenario

A	B	C	D	E	F	G	H
	Scenario Summary						
	Current Values:		200	300	400	500	
	Changing Cells:						
	\$O\$4	100	200	300	400	500	
	Result Cells:						
	\$O\$9	880	1760	2640	3520	4400	
Notes: Current Values column represents values of changing cells at time Scenario Summary Report was created. Changing cells for each scenario are highlighted in gray.							

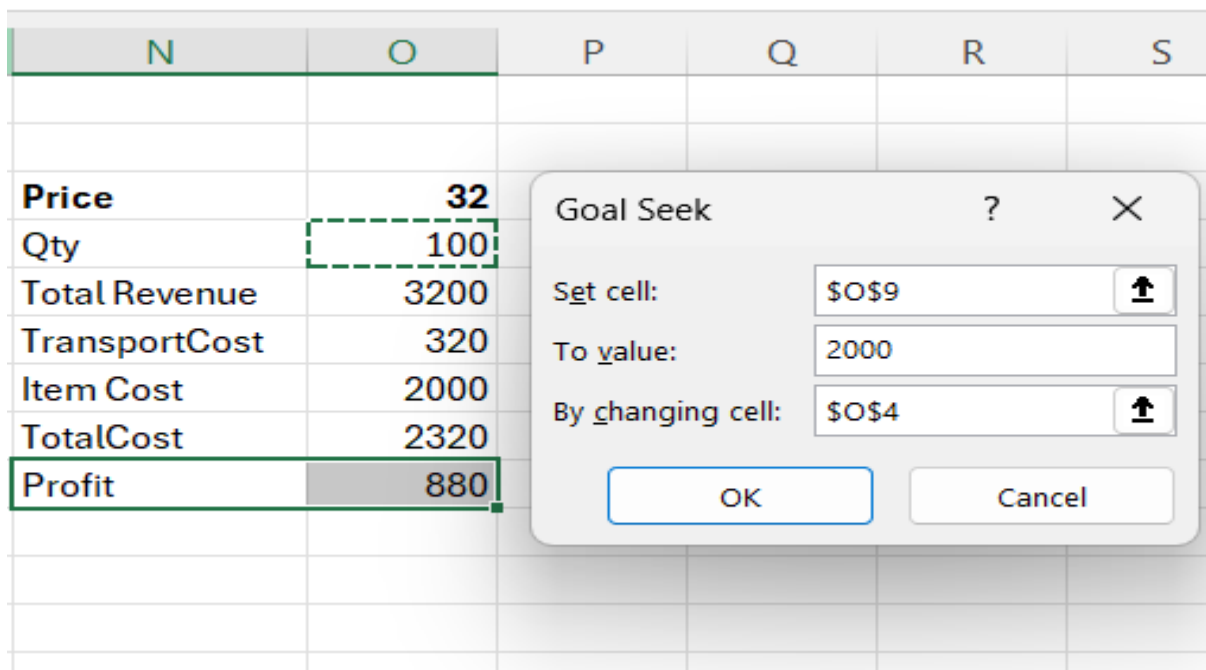
Goal seek:

Step 1: Select Profit cell and Go to Data tab click on **What-if Analysis**, Select **Goal seek**.



Step 2: In the goal seek dialog box,

- **Set cell:** Select the result cell (i.e. Profit).
- **To value:** Enter the target value (i.e. desired profit [2000])
- **By changing cell:** Select the input cell (i.e. Qty)



Step 3: Click OK. Excel Automatically calculates and displays the required input value. Click OK to apply the result.

	N	O	P	Q	R	S
Price		32				
Qty		227.2727				
Total Revenue		7272.727				
TransportCost		727.2727				
Item Cost		4545.455				
TotalCost		5272.727				
Profit		2000				

Goal Seek Status
 ?
 ✕

Goal Seeking with Cell O9 found a solution.

Target value: 2000

Current value: 2000

Step
 Pause
 OK
 Cancel

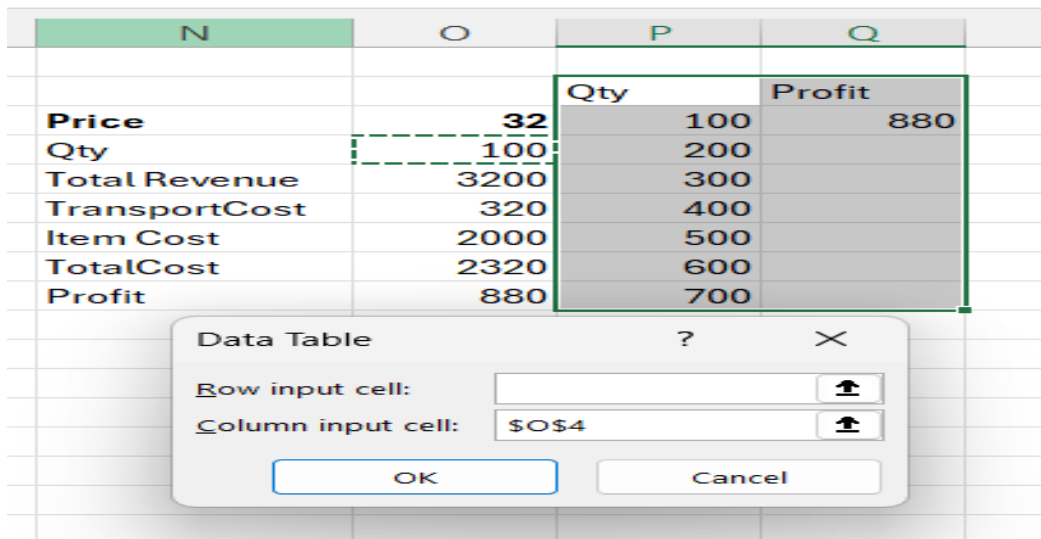
	N	O	P	Q	R
Price		32			
Qty		227.2727			
Total Revenue		7272.727			
TransportCost		727.2727			
Item Cost		4545.455			
TotalCost		5272.727			
Profit		2000			

Data table:

Step 1: Go to Data tab click on **What-if Analysis**, Select **Data table**.

Flash Fill	Remove Duplicates	Data Validation	Consolidate	Data Model	What-If Analysis	Forecast Sheet	Group
Data Tools					Scenario Manager...	Goal Seek...	Data Table...
	N	O	P	Q			
			Qty	Profit			
Price		32	100	880			
Qty		100	200				
Total Revenue		3200	300				
TransportCost		320	400				
Item Cost		2000	500				
TotalCost		2320	600				
Profit		880	700				

Step 2: In the dialog box, Enter the column input cell (the original input cell like Qty). Click OK.

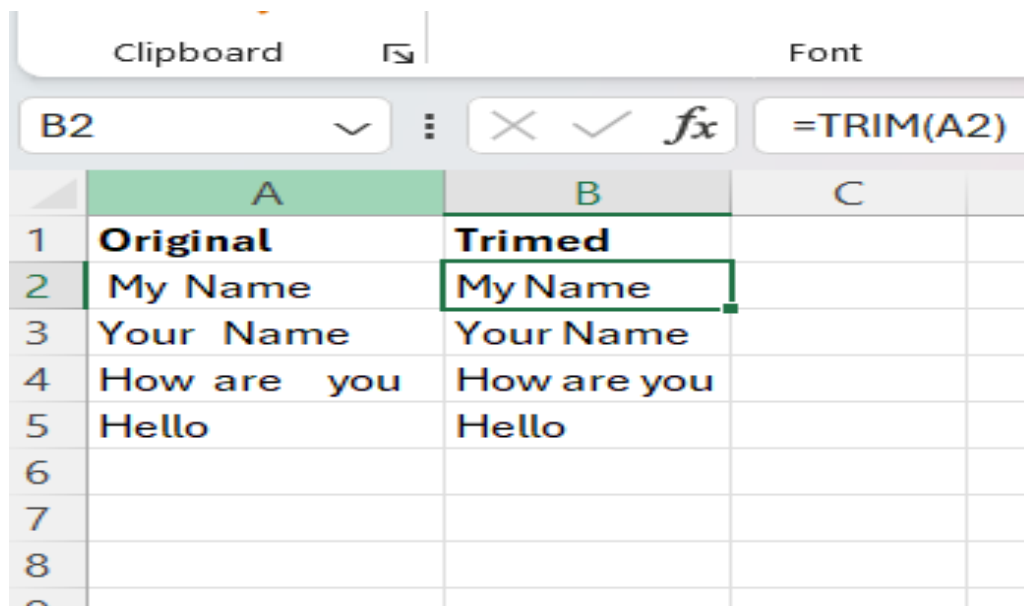


Excel automatically displays the result for each input value.

Data Tools				Forecast	
M	N	O	P	Q	R
			Qty		
	Price	32	100	880	
	Qty	100	200	1760	
	Total Revenue	3200	300	2640	
	TransportCost	320	400	3520	
	Item Cost	2000	500	4400	
	TotalCost	2320	600	5280	
	Profit	880	700	6160	

EXPERIMENT NO. 5: Cleaning data with text functions:

a. Use of TRIM function: Select the cell where you want to delete spaces> Use the formula =Trim(A2)>Click ENTER> Drag down to delete spaces of all required cell.

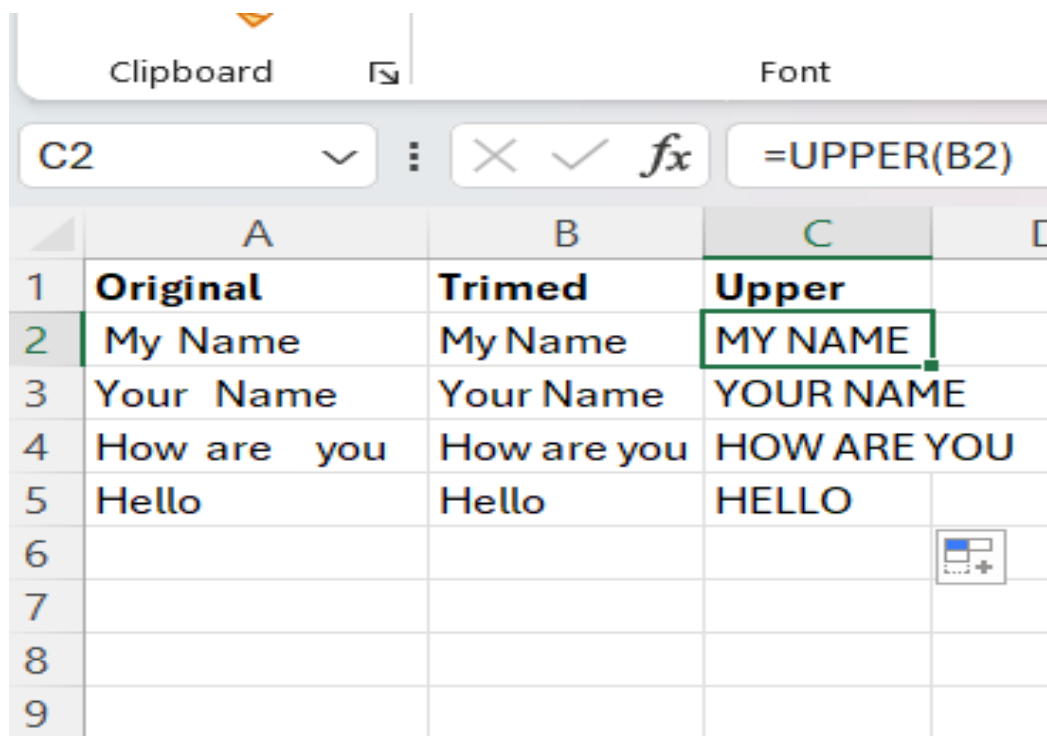


The screenshot shows an Excel spreadsheet with a ribbon at the top containing 'Clipboard' and 'Font' tabs. The formula bar at the top displays 'B2' and the formula '=TRIM(A2)'. The spreadsheet has three columns: A, B, and C. Column A is labeled 'Original' and contains the text 'My Name', 'Your Name', 'How are you', and 'Hello'. Column B is labeled 'Trimed' and contains the text 'My Name', 'Your Name', 'How are you', and 'Hello'. The cell B2 is selected, and the formula bar shows the formula '=TRIM(A2)'.

	A	B	C
1	Original	Trimed	
2	My Name	My Name	
3	Your Name	Your Name	
4	How are you	How are you	
5	Hello	Hello	
6			
7			
8			
9			

b. Upper; UPPER(B2)

Converts the selected contents of the cell to capital letters.



The screenshot shows an Excel spreadsheet with a ribbon at the top containing 'Clipboard' and 'Font' tabs. The formula bar at the top displays 'C2' and the formula '=UPPER(B2)'. The spreadsheet has four columns: A, B, C, and D. Column A is labeled 'Original' and contains the text 'My Name', 'Your Name', 'How are you', and 'Hello'. Column B is labeled 'Trimed' and contains the text 'My Name', 'Your Name', 'How are you', and 'Hello'. Column C is labeled 'Upper' and contains the text 'MY NAME', 'YOUR NAME', 'HOW ARE YOU', and 'HELLO'. The cell C2 is selected, and the formula bar shows the formula '=UPPER(B2)'.

	A	B	C	D
1	Original	Trimed	Upper	
2	My Name	My Name	MY NAME	
3	Your Name	Your Name	YOUR NAME	
4	How are you	How are you	HOW ARE YOU	
5	Hello	Hello	HELLO	
6				
7				
8				
9				

c. Lower: LOWER(B2)

Converts the selected contents of the cell to small letters.

Clipboard

Font

D2







=LOWER(B2)

	A	B	C	D
1	Original	Trimed	Upper	Lower
2	My Name	My Name	MY NAME	my name
3	Your Name	Your Name	YOUR NAME	your name
4	How are you	How are you	HOW ARE YOU	how are you
5	Hello	Hello	HELLO	hello
6				
7				
8				

d.Concat: CONCAT (A2,"",A3)

Adds the selected contents of the first cell and second cell and makes the new sentence.

Clipboard		Font	
B2		$\times$ $\checkmark$ f_x	=CONCAT(A2,"",A3)
	A	B	C
1	Original	Concate	
2	My Name	My Name Your Name	
3	Your Name		
4	How are you		
5	Hello		
6			
7			
8			
9			

EXPERIMENT NO. 6

Cleaning data containing date and time value: Use of DATEVALUE function, DATEADD function, DATEDIF function, TIMEVALUE function.

Create columns like date, month, year and insert values.

Create DOB column with: =DATE (C3, B3, A3) C3=year, B3=month, A3=date.

1.DATEVALUE:

Date in the form of code numbers.

First need to convert to text then use date value function.

- Date format: =TEXT(D3,"MM-DD-YYYY")
- Date value: =DATEVALUE(E3)

2.DATEADD:

Adding required number of months, days, years.

- Year add: =DATE(C3+4, B3, A3) C3=year, B3=month, A3=date
- Month add: =DATE (C3, B3+4, A3)
- Day add: =DATE(C3, B3, A3+4)

3.DATEDIF:

Used to calculate difference between 2 dates. Create today column and use =TODAY().

- =DATEDIF(D3, G3,"Y") → calculating total no. of years from DOB to today.
- =DATEDIF(D3, G3,"YM") → calculating total no. of months from DOB to today.
- =DATEDIF(D3,G3,"MD") → calculating total no. of days from DOB to today.

D3=DOB, G3=Today

4.TIMEVALUE: Time in the form of code numbers. Create hours, minutes, seconds columns.

Convert it into time format: =TIME(N3,O3,P3) N3=hours, O3=MIN, P3=SEC.

Convert to text: =TEXT(Q3,"HH:MM:SS") Q3=Time.

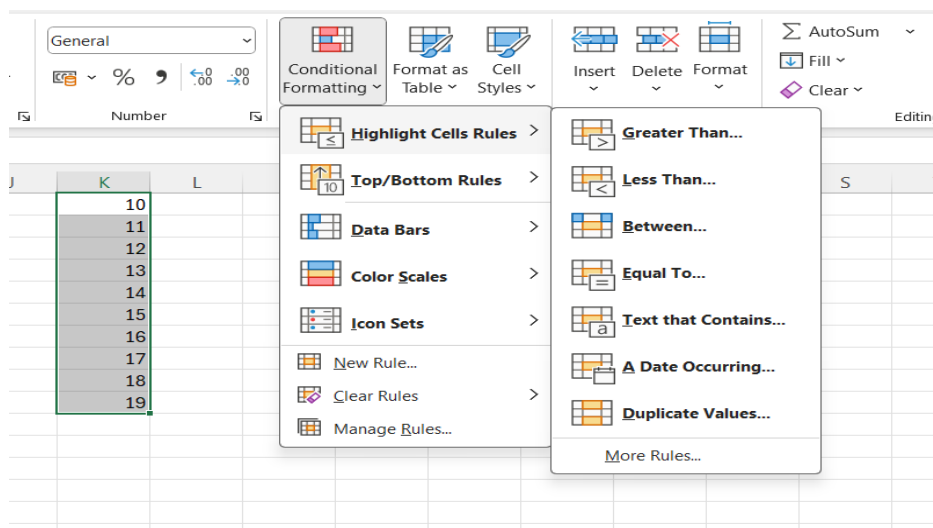
Time value: =TIMEVALUE(R3) R3=converted text of time

EXPERIMENT NO 7: Conditional formatting parsing, and highlighting data in spreadsheets during data analysis

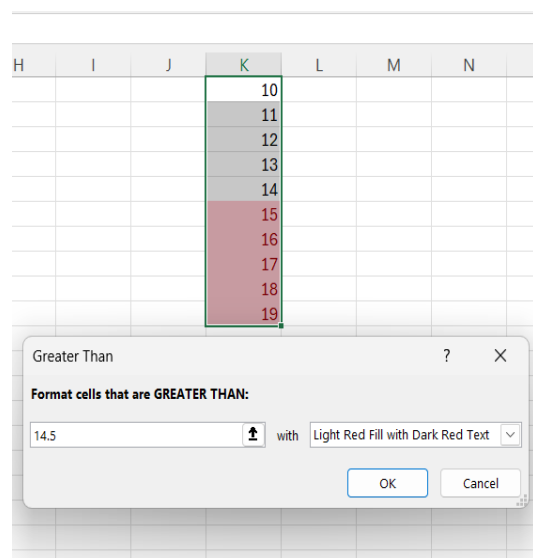
1.Conditional formatting:

Conditional formatting is a feature that helps you highlight data without manually formatting it. Using conditional formatting you can change cell colour, add data bars, add colour scales, add icons etc.

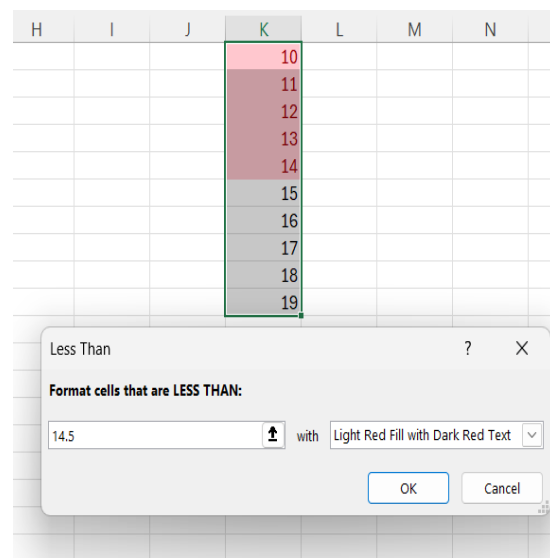
Steps: Select cells>Home>Conditional formatting (Highlight cells that have certain conditions)>Highlight cells with rules>



a. Greater than

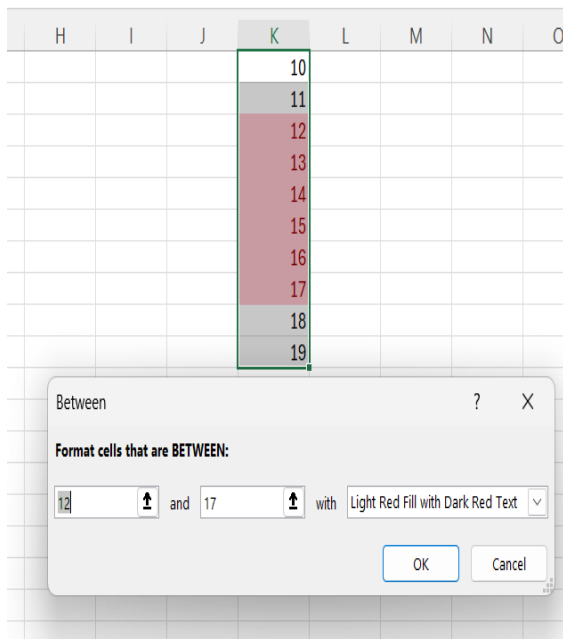


b. Lesser than

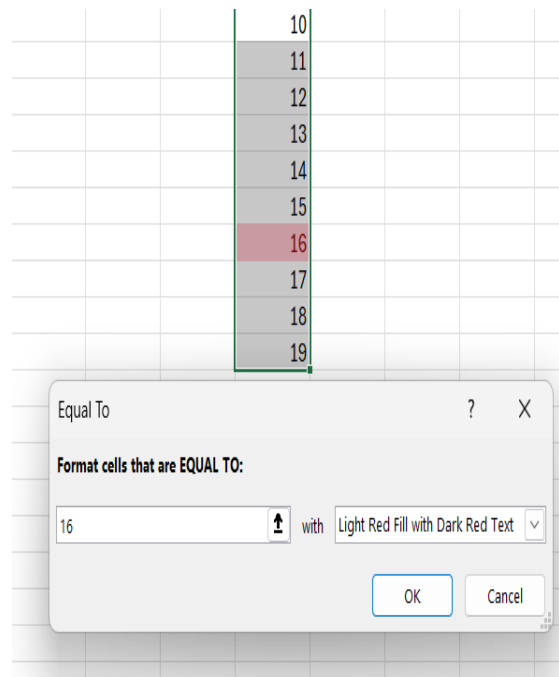


The highlighted cells in both greater than and lesser then shows the conditions given.

c. Between

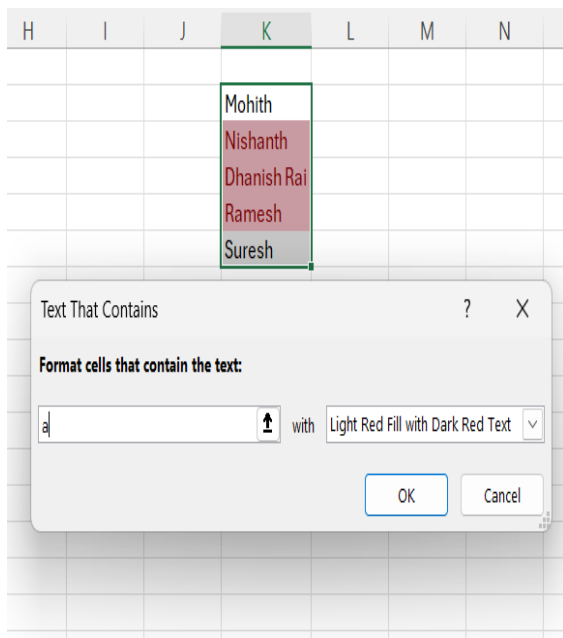


d. Equal to

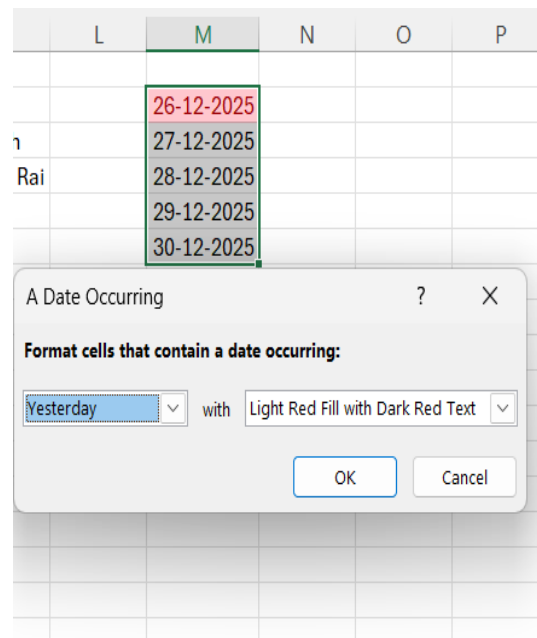


The “Between” function highlights the number specified in the range and “Equal to” highlights exact number given.

e. text that contains

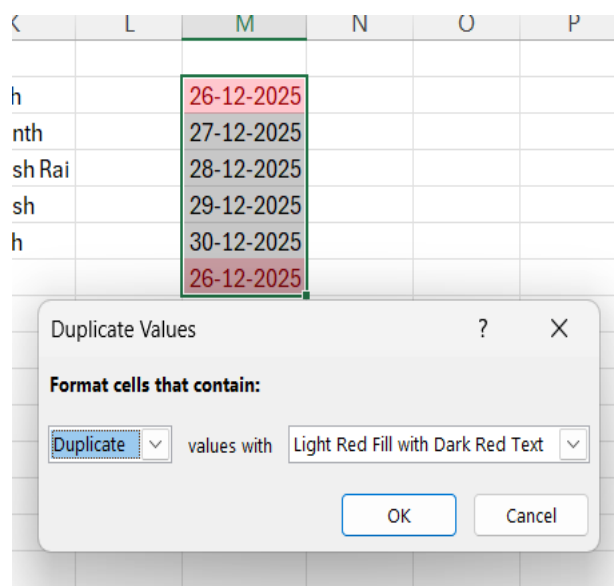


f. Date occurring



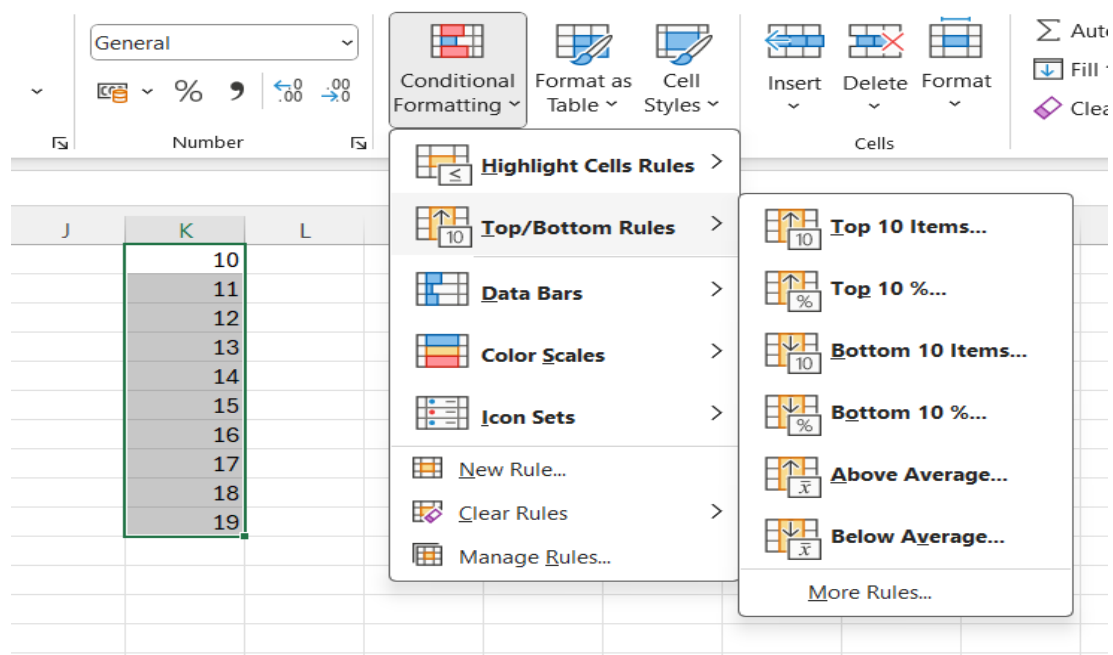
The “text that contains” highlights cells that contain the text given and “Date occurring” highlights the date specified.

g. Duplicate values:



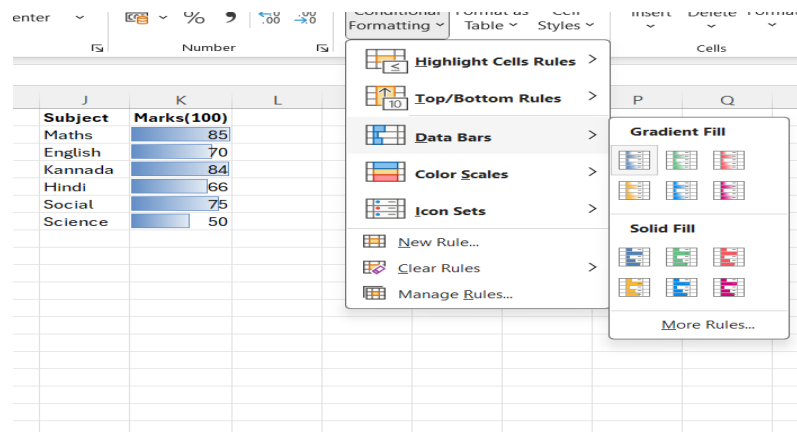
The “Duplicate values” highlights cells that contain the duplicate values in selected cells.

Top/Bottom cells



Data bars:

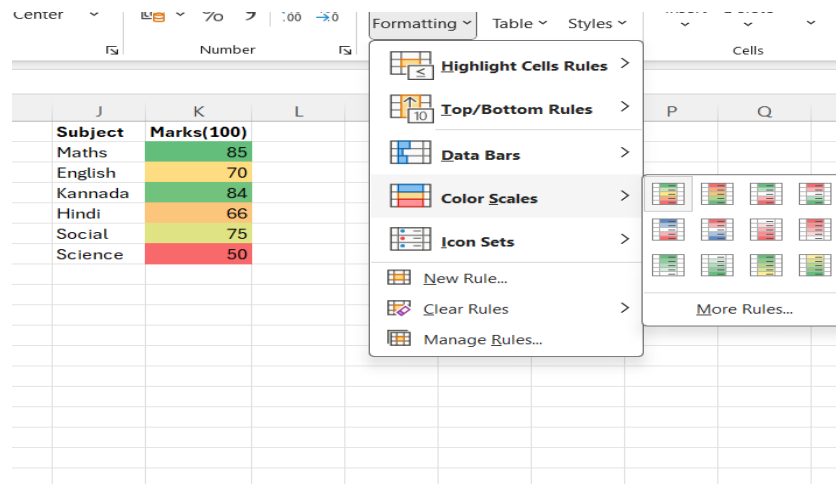
Select cells>Home>Conditional formatting> data bars



Colour

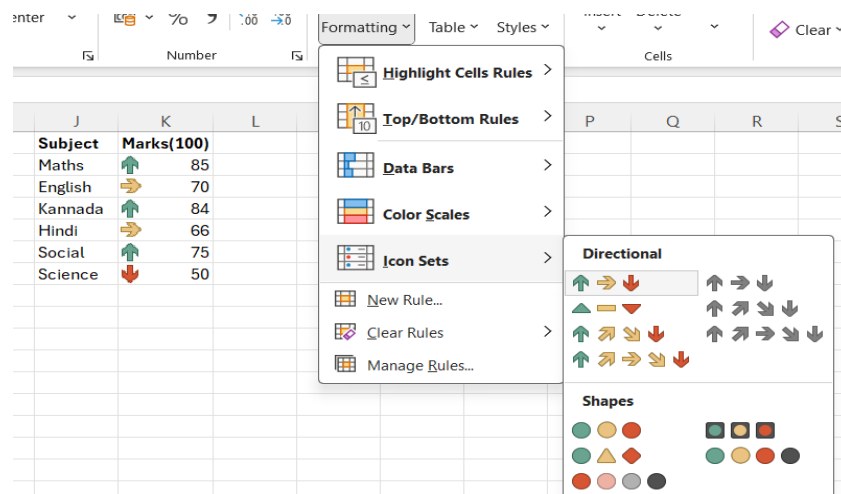
scales:

Select cells>Home>Conditional formatting> Colour scales



Icon Sets:

Select cells>Home>Conditional formatting> Icon sets

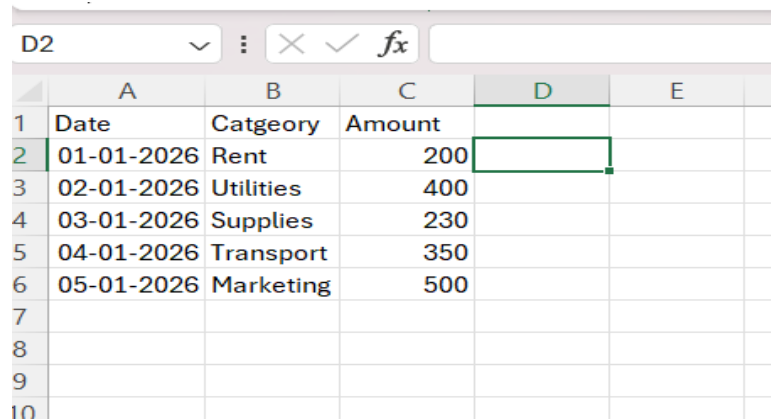


EXPERIMENT NO.8: Working with Multiple Sheets:

Work with multiple sheets within a workbook is crucial for organizing and managing data, perform complex calculations and create comprehensive reports.

Add Some data in 3 sheet as shown below.

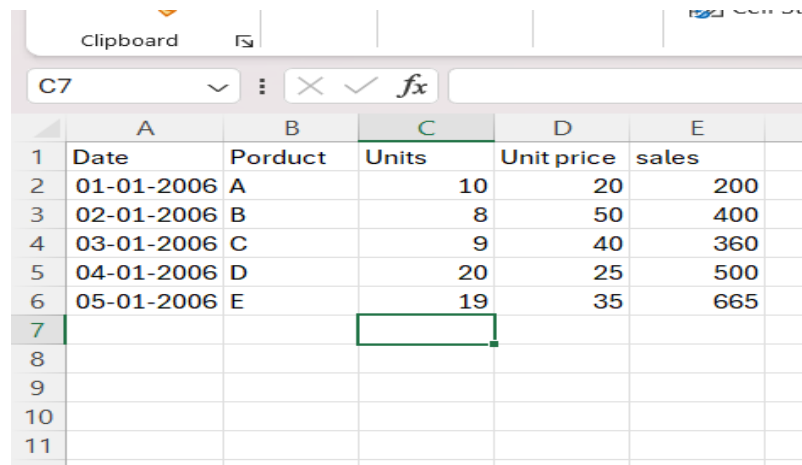
Expenses sheet



The screenshot shows an Excel spreadsheet with the following data:

	A	B	C	D	E
1	Date	Catgeory	Amount		
2	01-01-2026	Rent	200		
3	02-01-2026	Utilities	400		
4	03-01-2026	Supplies	230		
5	04-01-2026	Transport	350		
6	05-01-2026	Marketing	500		
7					
8					
9					
10					

Sales sheet



The screenshot shows an Excel spreadsheet with the following data:

	A	B	C	D	E
1	Date	Porduct	Units	Unit price	sales
2	01-01-2006	A	10	20	200
3	02-01-2006	B	8	50	400
4	03-01-2006	C	9	40	360
5	04-01-2006	D	20	25	500
6	05-01-2006	E	19	35	665
7					
8					
9					
10					
11					

Summary sheet

Total sales: =sum(sales!E2:sales!E6)

Total expenses: = sum(Expenses!C2:Expenses!C6)

Profit: =B1-B2

Max sales in a day: =max(sales!E2:sales!E6)

Min sales in a day: =min(sales!E2:sales!E6)

Then go to View then New window. Copy of the window is created. Again go to view click on “Arrange all” then arrange window horizontally, vertically etc.

Clipboard

B1 ✕ ✓ *fx* =SUM(sales!E2:sales!E6)

	A	B	C	D	E
1	TOTAL SALES	2125			
2	TOTAL EXPENSES	1680			
3	PROFIT	445			
4	MAX SALE IN ADAY	665			
5	MIN SALE IN A DAY	200			
6					
7					
8					

AutoSave on

Myexcel - 2 - Excel

File Home Insert Draw Page Layout Formulas Data Review View Help Acrobat

Sheet View Workbook Views Show Zoom 100% Zoom to Selection New Window Arrange All Freeze Panes Window Switch Windows Macros

	A	B	C	D	E	F	G	H	I	J	K
1	Date	Category	Amount								
2	01-01-2026	Rent	200								
3	02-01-2026	Utilities	400								
4	03-01-2026	Supplies	230								
5	04-01-2026	Transport	350								
6	05-01-2026	Marketing	500								

Sheet27 Expenses sales summi

AutoSave on

Myexcel - 1 - Excel

File Home Insert Draw Page Layout Formulas Data Review View Help Acrobat

Sheet View Workbook Views Show Zoom 100% Zoom to Selection New Window Arrange All Freeze Panes Window Switch Windows Macros

	A	B	C	D	E	F	G	H	I	J
1	TOTAL SALES	2125								
2	TOTAL EXPENSES	1680								
3	PROFIT	445								
4	MAX SALE IN ADAY	665								
5	MIN SALE IN A DAY	200								

Sheet27 Expenses sales summi

EXPERIMENT NO.9:

Create worksheet with following fields: Emp no, E name, Basic Pay(BP), Travelling Allowance(TA), Dearness Allowance(DA), House Rent Allowance(HRA), Income Tax(IT), Provident Fund(PF), Net Pay(NP). Use appropriate formulas to calculate the above scenario. Analyse the data using appropriate chart and report the data.

Emp no → (A1), E name → (B1), Basic Pay (BP) → (C1)

Travelling Allowance (TA): =IF(C2>=20000, C2*20%, IF(C2>=15000, C2*15%, C2*10%)) →(D1)

Dearness Allowance (DA): =IF(C2>=20000, C2*25%, IF(C2>=15000, C2*20%, C2*15%)) →(E1)

House Rent Allowance (HRA): =C2*10% → (F1)

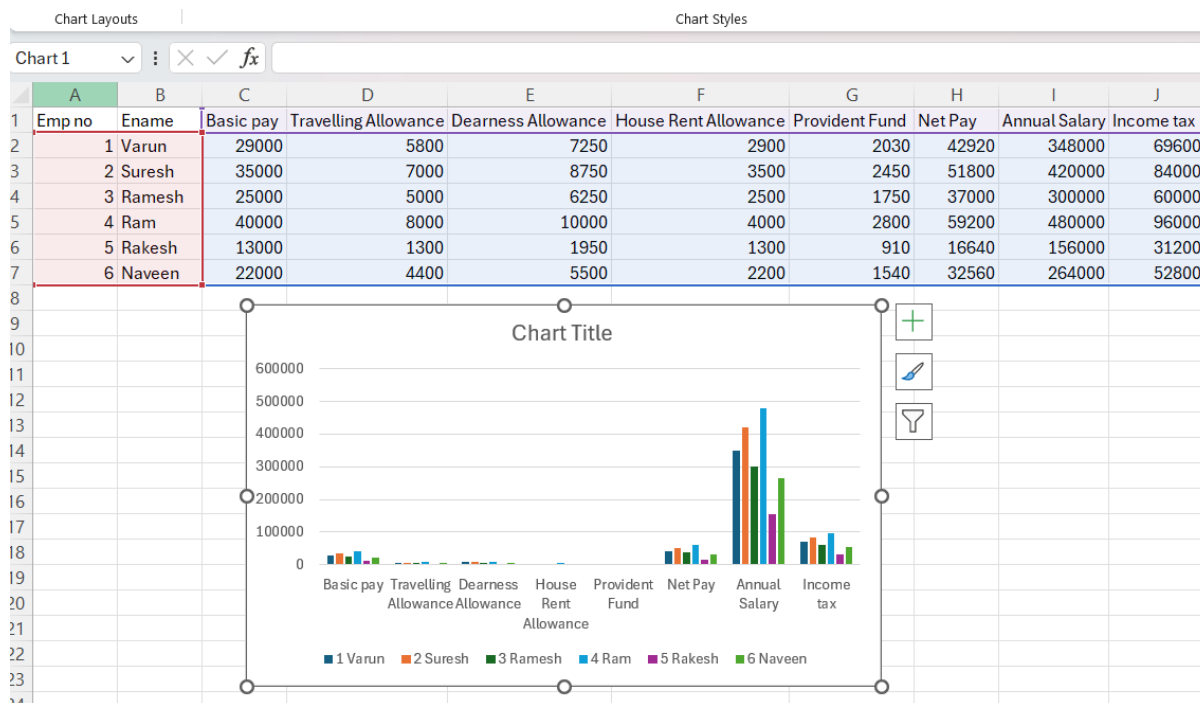
Provident Fund (PF): =C2*7% → (G1)

Net Pay (NP): =C2+D2+E2+F2-G2 →(H1)

Annual Salary: =C2*12 → (I1)

Income Tax: =IF(I2>50000, I2*20%, I2*10%) →J1

Add Chart: Select entire table then go to Insert → 2D column chart.



EXPERIMENT NO.10:

Create worksheet on Inventory Management: Sheet should contain Product code, Product name, Product type, MRP, Cost after % of discount, Date of purchase. Use appropriate formulas to calculate the above scenario. Analyse the data using appropriate chart and report the data.

Product code → (A1)

Product name → (B1)

Product type → (C1)

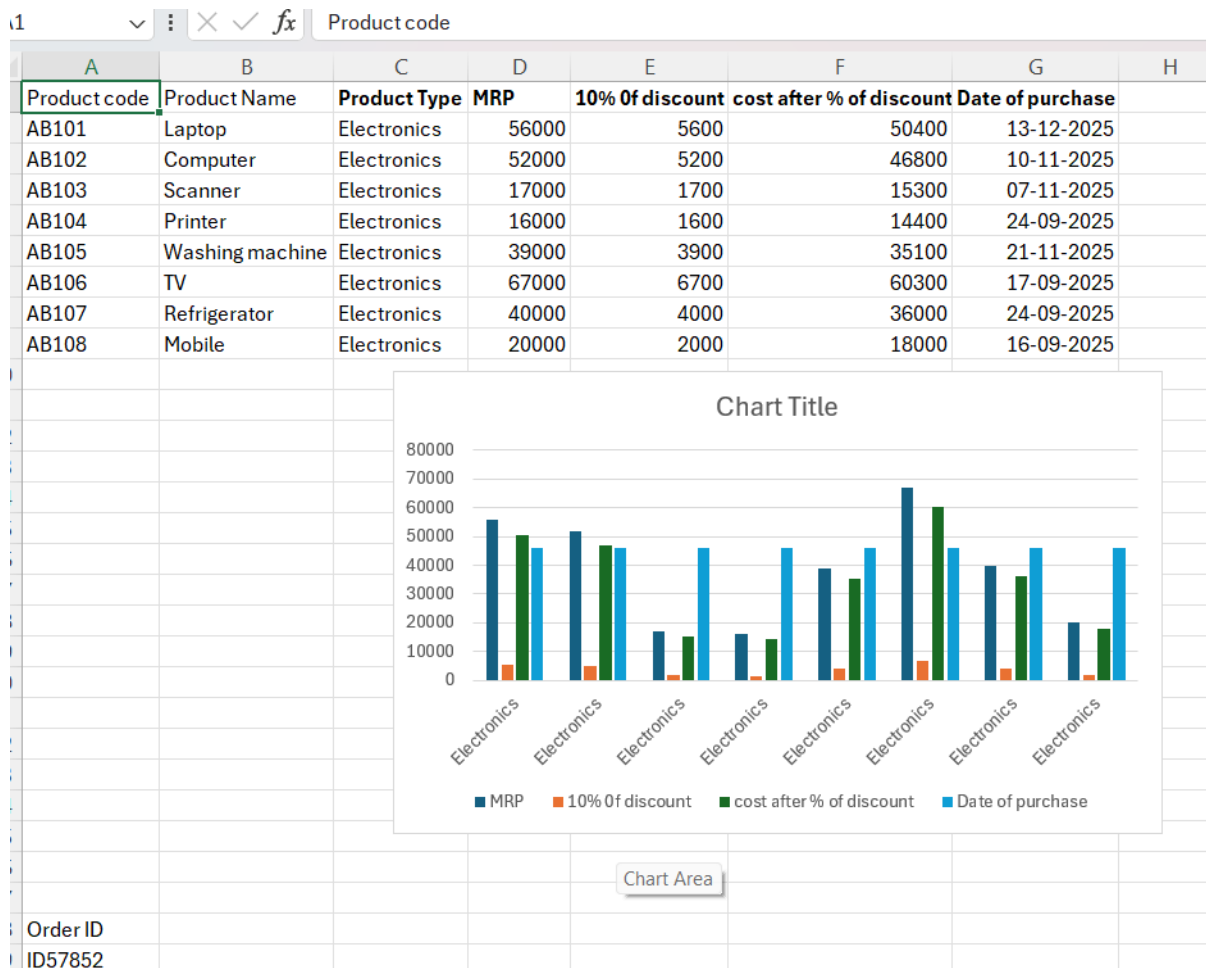
MRP → (D1)

10% of discount: =D2*10/100 → (E1)

Cost after % of discount: =D2-E2 → (F1)

Date of purchase Add chart → (G1)

Add Chart: Select entire table then go to Insert → 2D column chart



EXPERIMENT NO 11:

Create worksheet on Sales analysis of Merchandise Store: data consisting of Order ID, Customer ID, Gender, age, date of order, month, online platform, Category of product, size, quantity, amount, shipping city and other details. Use of formula to segregate different categories and perform a comparative study using pivot tables and different sort of charts.

Step1: Create a table where the columns are:

Order ID, Customer ID, Gender, Age, date of order, month, online platform, Category of product, size, quantity, rate, total amount(=Quantity*Rate), shipping city

Step 2: Insert> Pivot table> select location>ok

Step 3:

- In filters, shipping city, age
- In columns, online platform
- In rows, Category of product, Date of order
- In values, sum of total amount

Step 4: Pivot chart and table for same data

M12

X

✓

f_x

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Order ID+	Customer ID	Gender	Age	Date-of-order	Category o	Online Platf	Size	Quantity	Rate	Total amount	Shipping city	
2	ID57852	CID745896	MALE	35	12-07-2025	MOBILE	AMAZON	L		1	20000	20000	MYSORE
3	ID57853	CID745897	FEMALE	25	13-07-2025	FOOTWEA	FLIPCART	M		1	2000	2000	HASSAN
4	ID57854	CID745898	FEMALE	25	14-07-2025	JEANS	AJIO	L		2	1500	3000	BANGLORE
5	ID57855	CID745899	MALE	20	15-07-2025	GOWN	MEESHO	S		4	2000	8000	BELLUR
6	ID57856	CID745900	MALE	45	16-07-2025	LIPSTICK	BLINKIT	L		2	700	1400	GOA
7	ID57857	CID745901	FEMALE	42	17-07-2025	KAJAL	INSTAMAR	M		8	500	4000	KERALA
8	ID57858	CID745902	MALE	28	18-07-2025	BABY SHO	JIO	S		5	450	2250	UTTAR PRADESH
9	ID57859	CID745903	MALE	22	19-07-2025	SAREE	MYNTRA	M		1	2500	2500	PUNE
10	ID57860	CID745904	FEMALE	32	20-07-2025	EAR PHON	NYKA	L		2	2000	4000	SURTHKKAL
11													

EXPERIMENT NO.12: Generation of report & presentation using Auto filter ¯o.

Macros: A Macro is a recorded or written set of instructions (VBA code) that automates repetitive tasks in Excel.

Step 1: Create table of USN, Name, Year.

Go to View then Macros then click Record Macro. Give Macro name then ctrl+ shift+ K.

Apply Filter: go to Data then Filter, add chart, fill colours or make any changes

The screenshot shows the Microsoft Excel interface. The 'Macros' menu is open, displaying options: 'View Macros', 'Record Macro...', and 'Use Relative References'. The 'Record Macro...' option is highlighted. Below the menu, a data table is visible with columns A, B, and C. The table contains the following data:

	A	B	C
1	Name	Age	Year
2	Ram	21	2004
3	Rakesh	23	2001
4	Rajesh	20	2005
5			
6			
7			
8			
9			
10			
11			
12			
13			

Step 2: Now go to View then Macros then click stop record.

Create new table in new sheet with same number of rows and column.

Select full table then go to view then macros then view macros there select required macros then click on RUN.

Whatever the changes made in sheet 1 will be copied to sheet2.

Sheet View			Workbook Views				
A1	✕	✓	<i>fx</i>	Name			
	A	B	C	D	E	F	
1	Name	Age	Year				
2	Mohith	20	2005				
3	Nishanth	19	2006				
4	Dhanish	19	2006				
5							
6							
7							
8							
9							
10							
11							
12							
13							
14							